Warehouse Order-Picking Optimisation

Re-slotting and re-routing a footwear manufacturer's warehouse using its own 2023 picking data

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-59.5%
walking per wave
(166 m to 67 m)

-26.3%
time per wave
(7.8 to 5.8 min)

+36%
units per picker-hour
(168 to 229)

16%
of stock moved
(186 locations)

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1. Executive summary

A Canadian safety-boot manufacturer runs a single-depot, 4-level warehouse in Sherbrooke (Quebec) under a **random storage** policy with **mixed shelving**: up to 18 different products share one location, pickers walk with a trolley of at most 27 items, and the WMS releases picking waves in order-arrival sequence without any optimisation. The case asks whether a different storage arrangement and better routing can shorten the time between an order being received and its items arriving at the single I/O (exit) point.

To answer it, I built a Python model (*warehouse_optimization.py*) that turns the nine case files into a digital twin of the warehouse. The model rebuilds the rack as a walking network in real metres, then **replays all 9,707 picking waves of 2023** (215,192 units, 22 operators) under seven storage arrangements and three routing methods. Every arrangement is judged on exactly the same work, so the comparison is fair. The new arrangement was designed only on the first 70% of the year and then tested on the last 30%, which the design never saw.

Measure (7,341 fully comparable waves)	AS-IS today	Best case policy (Hybrid)	Proposed (Optimised-FAMILY)
Walking per wave	166 m	122 m (-26.7%)	**67 m (-59.5%)**
Time per wave	7.85 min	7.02 min (-10.5%)	**5.78 min (-26.3%)**
Units per picker-hour	168	188	**229 (+36%)**
Walking share of picker time	35%	29%	19%
90th-percentile walk per wave	272 m	211 m	105 m
Saving on unseen test waves	-	-26.3%	**-58.8%**

Table 1. *Headline results. Percentages are against AS-IS (recorded WMS sequence). 'Best case policy' is the best of the four storage policies supplied with the case.*

Key findings

- **Routing alone is a free win.** Re-ordering today's pick lists with a nearest-neighbour + 2-opt algorithm cuts walking by 14.6% (166 to 142 m) without moving a single box.
- **Slotting is the big lever.** Putting fast movers in a compact forward zone around the I/O removes another 44.9 percentage points of walking. Together: -59.5% walking and -26.3% time per wave.
- **The result is robust.** It holds on unseen test waves (-58.8%), 97% of individual waves improve (Wilcoxon $p < 0.001$), and the time saving stays between 20% and 32% across all tested walking speeds and pick times.
- **Of the four case policies, Hybrid is best** (122 m) and Dedicated worst (163 m); the proposed design beats Hybrid on 85% of waves.
- **Moving 16% of stock is enough.** Only the 186 locations nearest the I/O are re-slotted (6,738 unit moves); 3,818 of those moves concern class-A items, which generate 80% of pick visits, so they come first.
- **The data revealed operational issues:** the master ABC classes match real demand for only 56% of products, the whole lower rack block (walkways W01-W07) recorded zero picks in 2023, about 11% of picks are logged at a corridor point (RC-01) rather than a shelf, and workload is very unevenly spread across operators (Gini 0.53).

Recommendations (detailed in Section 13)

- Phase 0 (this week): switch WMS pick-list sequencing to the optimised route order.
- Phase 1 (a few days, 64-127 labour hours): re-slot the class-A forward zone around the I/O (3,818 moves).
- Phase 2: complete the forward zone (remaining 2,920 moves) and introduce daily replenishment of it.
- Refresh the ABC master quarterly from WMS data, clarify what RC-01 is, and balance waves across operators.

> **Scope of the numbers**
> All results in this report come from the model run of 10 October 2026 with the configuration listed in Appendix A. The model is a *static* evaluation: each pick is assigned to the nearest shelf that holds the item, and the forward zone holds two days of average demand. Section 14 explains what this assumption means and how the final version of the code tests it more strictly.

2. The case

2.1 Company and warehouse

The dataset comes from the warehouse of a footwear manufacturer that makes high-quality boots for the construction, mining and manufacturing sectors. Each product is a box holding one pair of boots and is identified by a reference code plus a size. The warehouse is semi-automated: storage is in racks, but picking is manual, with operators pushing a trolley through the aisles and bringing the collected items to a single input/output (I/O) point, from where they move to the staging area and the three shipping bays.

Parameter	Value (from the case article)
Storage policy in use	Random storage, mixed shelving (up to 18 different products per location)
Levels	4 levels. Levels 1-2 (1.03 m high each) are hand-picked, 3 compartments per bay, 612 kg per level. Levels 3-4 (1.79 m high each) are replenishment levels served by forklift, 1,293 kg per position.
Racks and aisles	25 racks, 17 parallel aisles, 3 cross ('vertical') aisles: left (LC), central (CC), right (RC)
Storage locations	2,293 (2,292 with coordinates in the data)
Shelving unit (bay)	3.66 m long x 1.52 m deep
Boxes	Pick face: small box 0.381 x 0.381 x 0.152 m. Reserve: large box 0.787 x 0.470 x 0.416 m (6 small boxes)
Product weights	1.2 kg (light), 1.8 kg (standard), up to 2.6 kg (heavy); not given per product
Pick list / trolley	15 to 27 products per pick list; trolley capacity 27 items
Depot	Single I/O point; WMS builds waves in order-arrival sequence with no batching or route optimisation

Table 2. *Warehouse facts used in the model.*

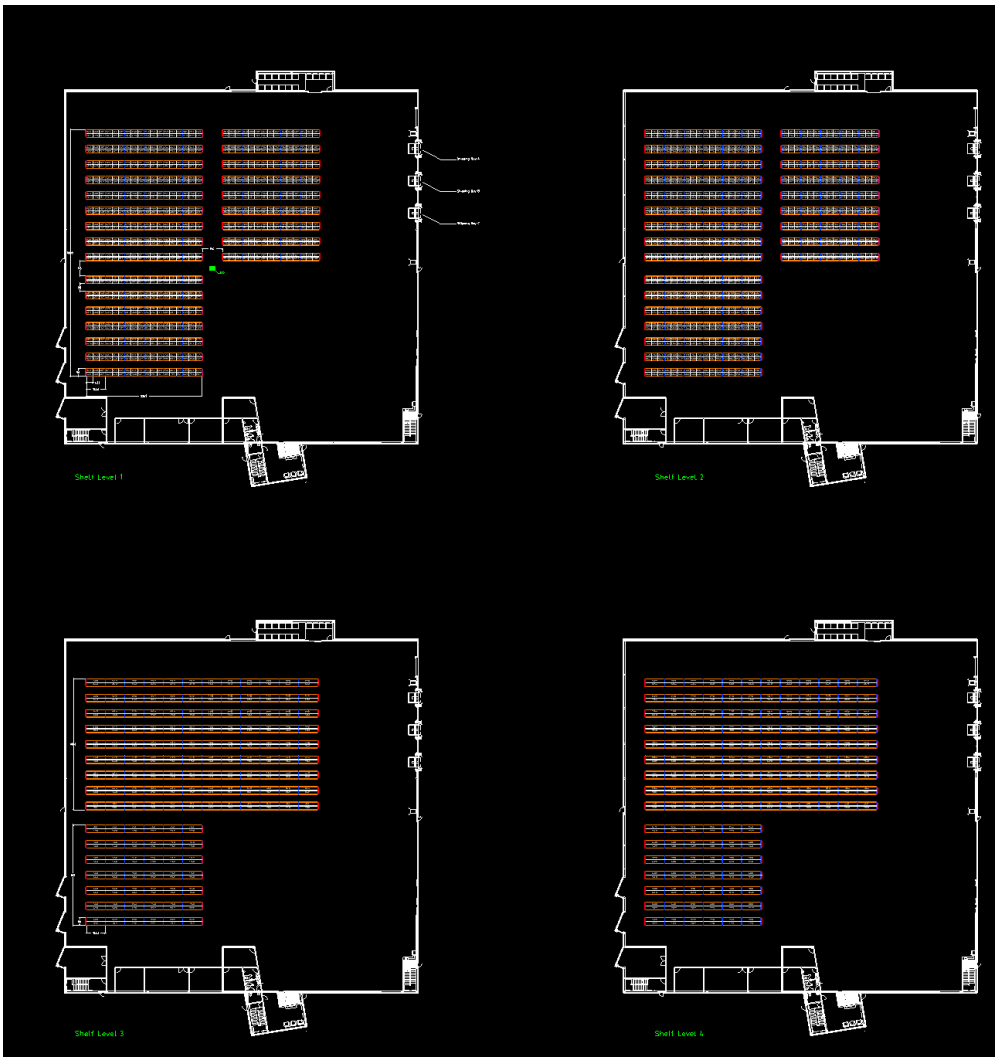


Figure 1. *Case CAD drawing of the four shelf levels. Levels 1-2 (top) contain the hand-picking positions; levels 3-4 (bottom) hold one wide replenishment position per bay. The three shipping bays are on the right wall.*

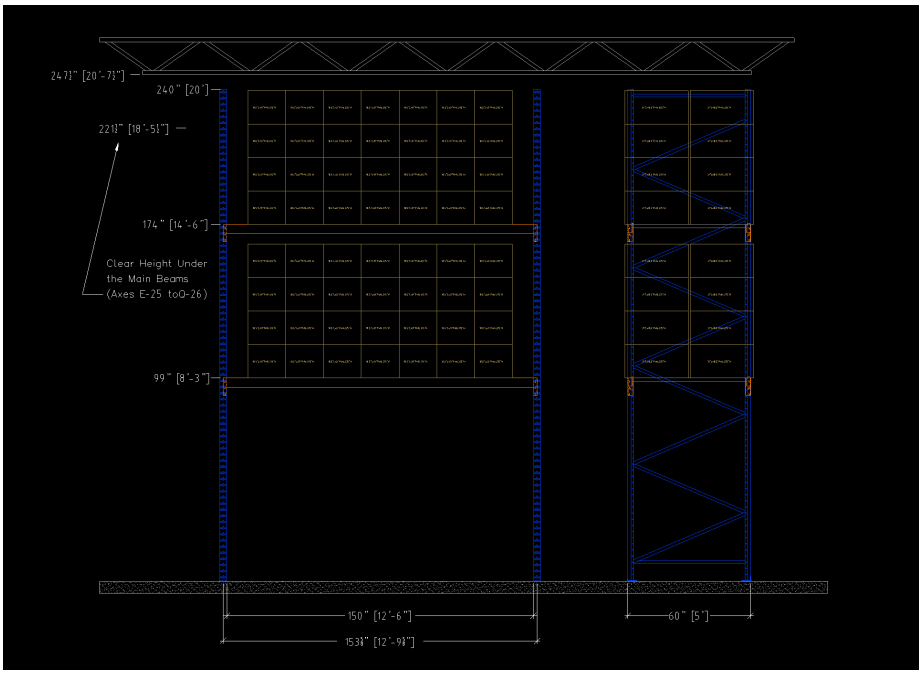


Figure 2. *Case CAD elevation of one rack: beams at 99 in (8 ft 3 in) and 174 in (14 ft 6 in), 150 in clear span and 60 in (1.52 m) depth. The lower part is the hand-pick zone; the stacked large boxes above are reserve stock.*

2.2 Problem statement and objectives

Under random storage, a popular product can sit anywhere in the building, so the pickers' walk is long and unpredictable. Travel is typically the largest single component of manual picking time in the warehousing literature (de Koster et al., 2007), which makes it the natural target. The objective set for this work was to reduce the time from order receipt until the order's items reach the I/O point, through five mandatory deliverables:

- Exploratory data analysis (EDA) of all files.
- Walking distance and time for every storage arrangement supplied with the case, plus a new arrangement that needs less time.
- An Excel description of the new arrangement (the re-shuffle) and the reconstructed picking routes.
- New warehouse maps and visualisations, including demand and ABC views.
- Picking performance per arrangement and per operator.

2.3 The dataset

The case provides nine CSV files extracted from the company's WMS, plus CAD drawings and a Python notebook that only draws the layout. Two different delimiters are used (semicolon and comma), and some cells contain a semicolon inside them (for example *TQBVRI;7* = reference;size), which is why a dedicated loader was needed.

File	Rows	What it contains
Product.csv	208	Product master: reference, ABC class (ABCCOD), sector
Customer_Order.csv	122,370	Order lines: customer, order, reference, size, quantity, creation date, wave, operator
Picking_Wave.csv	215,192	One row per unit picked: wave, reference, size, location, operator (recorded order)
Storage_Location.csv	2,292	Location code and x, y, z coordinates
Support_Points_Navigation.csv	44	Corridor points LC-01..17, CC-01..17, RC-08..17
Random_Storage.csv	2,292	Stock in every location under the random policy (18 slots each)
Class_Based_Storage.csv	2,341	Stock under ABC class-based storage
Dedicated_Storage.csv	2,340	Stock under dedicated (fixed home) storage
Hybrid_Storage.csv	2,340	Stock under hybrid storage (20% dedicated near I/O, rest random by class)

Table 3. *The nine input files.*

Location codes follow the pattern *J-10-21*: rack face J, column 10, level 2, position 1. Levels 3 and 4 have a single position per bay (codes ending in -30 and -40).

3. Approach and the role of the code

3.1 Why a replay (digital-twin) approach

There are no pick timestamps in the data, so the company's real picking times cannot be measured directly. What the data does contain is the exact composition of every wave, the location each unit was taken from, the order in which the WMS listed the picks, and the full stock picture under four alternative policies. That is enough to **replay** history: for each wave, place its items according to a given arrangement, plan the picker's walk through the real aisle network, and convert distance and work content into time. Because every arrangement processes the identical 2023 workload, differences in the results can only come from the arrangement and the routing, not from demand.

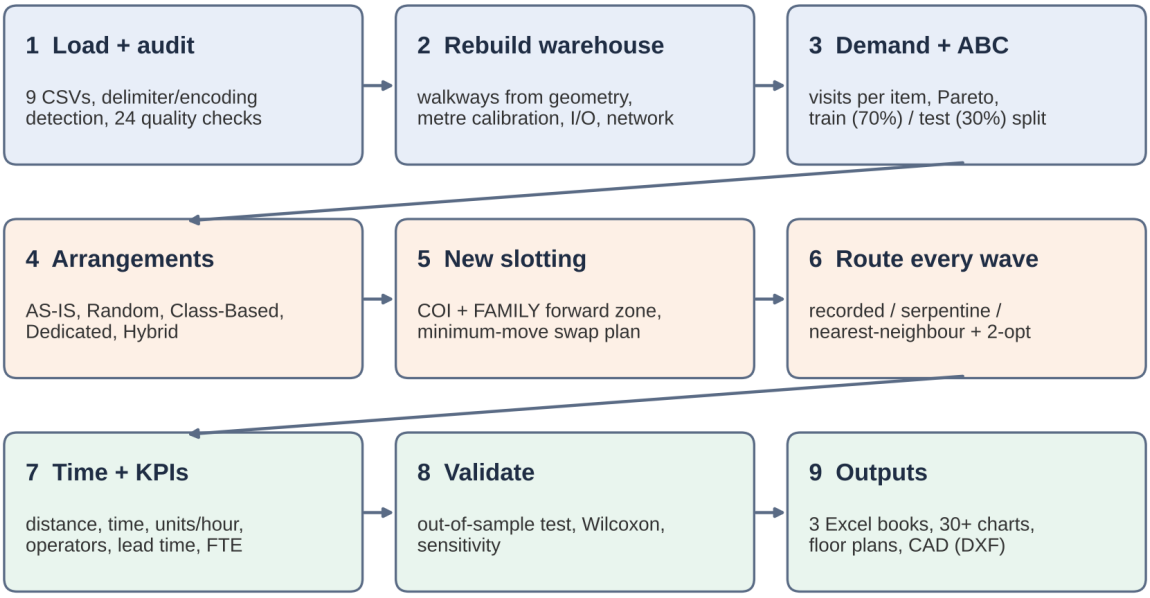


Figure 3. *The nine stages of the analysis pipeline, each implemented as functions in warehouse_optimization.py.*

3.2 What each part of the code does

The code is a single Python file of about 2,900 lines, organised as a pipeline. Table 4 maps each stage to the functions that implement it and to the problem it solves.

Stage	Main functions	What it does / why it was needed
1 Load	locate_and_load, read_table, load_storage_location	Finds the 9 files (folder, Drive, zip or upload), detects delimiter and encoding per file, normalises line endings, reads coordinates even when the 'position' field contains commas.
1 Clean	clean_tables, storage_to_long, norm_size	Builds clean tables, splits 'REF;SIZE' cells, converts 85 to 8.5, keeps duplicate pick rows (each is one unit), runs 24 data-quality checks.
2 Geometry	build_geometry, Network	Finds which axis runs along the aisles, assigns every rack face to its walkway from the coordinates, calibrates metres per axis, places the I/O, builds the distance network.
3 Demand	build_demand, abc_from_counts	Counts pick visits and units per item, ABC classes, Pareto curve, chronological 70/30 train/test split.
4 Policies	inventory_with_ids, assign_pick_locations	For each case policy, decides where each pick would be taken from (nearest pick-face slot holding the item).
5 Slotting	design_slotting, min_move_layout	Designs the new forward zone (COI and FAMILY variants) and converts it into a minimum-move swap plan.
6 Routing	tour_nn_2opt, tour_serpentine, evaluate_scenarios	Plans the walk for every wave under every arrangement and routing method; caches identical routes.
7 KPIs	scenario_summary, time_variation	Converts distance into time, computes distance, time, productivity, operator, lead-time and staffing KPIs.
8 Validation	(in main) wilcoxon, sensitivity loop	Out-of-sample test, paired statistical tests, sensitivity of the time saving to the assumptions.
9 Outputs	draw_frame, heat_layer, draw_level_plans, export_cad, write_xlsx	Maps, charts, level-by-level floor plans (PDF/PNG), CAD files (DXF), three Excel workbooks, summary.

Table 4. *Code structure and role of each stage.*

The following sections walk through these stages in order, showing for each the analysis performed, the charts it produced and what they mean.

4. Data preparation and quality audit

Before any analysis, the code audits every file and records each issue together with the action taken in the *Data_Quality* sheet. Nothing is silently deleted: the rule throughout was to keep, repair or flag, and to exclude only what could not be placed on the map.

Check	Count	Meaning and action taken
Exact duplicate rows in Picking_Wave	102,201	Every row has quantity 1, so duplicates are separate units of the same item. **Kept**; removing them would have cut the workload by almost half.
Sizes written without the decimal point	41,549	Customer_Order uses 85 for 8.5. Values above 20 were divided by 10.
Missing size in orders	16	Kept; matched at reference level.
Waves above the 27-item trolley capacity	74	Maximum observed 30 items. Kept and flagged.
Waves in orders but not in picking	77	No pick data; excluded from routing.
Waves whose ordered units differ from picked units	2,313 of 9,707	Informational (partial picks or order edits).
References in Class_Based_Storage not in Product.csv	109	Kept as stock, since they occupy space.
Location K-23-40 stored with level 3 and same x, y, z as K-23-30	2	Level taken from the code; flagged.
Locations used in policies/picks but missing coordinates	65	47 imputed from neighbouring positions of the same bay; 18 unresolved.
Historical picks at a location that is not a shelf	23,559	RC-01 alone has 22,979 picks (10.7% of all units), plus RC-04, RC-05 and codes such as C-102. RC-01 is a corridor support point, probably a floor or bulk area. Excluded from AS-IS routing.
Picks whose exact reference+size exists in Random storage	88.7%	High enough to evaluate at SKU (reference+size) level; the remaining 11.3% use the nearest box of the same reference.

Table 5. *Main data-quality findings (full list in 01_Analysis_Tables.xlsx, sheet Data_Quality).*

Two consequences follow. First, because some waves contain picks that cannot be placed (mainly RC-01), the comparison between arrangements uses the **7,341 waves that are fully placed in every scenario** (76% of waves, 161,771 units). Comparing on the same waves avoids flattering any arrangement. Second, the RC-01 finding is itself a recommendation: a tenth of all picks happen at a location that is not part of the rack model, and the company should confirm what it is.

5. Rebuilding the warehouse as a walking network

The case coordinates are anonymised and do not come with units, and the original notebook only draws boxes. To measure walking distance, the code had to turn the coordinates into a physically meaningful network. Three problems had to be solved.

Problem 1: the letter is not always the walkway

On the case drawing, walkway LC-07 is faced by shelves labelled G (below) and H (above), and the right-hand block is lettered one letter lower than the left block (walkway 16 is faced by Q on the left but P on the right). Treating each letter as an aisle would route a picker from G to H the long way round. The code therefore assigns walkways from **geometry**: a rack face is defined by its letter, column parity (odd/even) and block (left/right of the central corridor). Each face opens onto the walkway on the side away from its rack partner (the other face of the same double-sided rack). The run confirmed that 11 of the 17 walkways are shared by two letters (for example W07 = G/H, W09 = I/J).

Problem 2: the coordinates have no unit, and the two axes use different scales

The code measures, in the data, the distance between consecutive bays along a walkway (50 units) and between neighbouring walkways (90 units), and matches them to the case dimensions: a bay is 3.66 m long and the CAD shows a walkway pitch of about 2.90 m (56 in rack + 58 in aisle). This gives 0.0732 m per unit along the walkways and 0.0322 m per unit across them. The rebuilt racked area measures **42.5 m x 46.4 m**, consistent with the CAD drawing, which validates the calibration.

Problem 3: pickers cannot walk through racks

Distances are therefore **rectilinear along the aisle network**, the standard model for parallel-aisle warehouses (Ratliff & Rosenthal, 1983). For two locations with along-walkway positions s_1, s_2 and walkway centre-lines c_1, c_2:

```
same walkway: d = |s1 - s2|
different walkways: d = min over cross aisles X of |s1 - X| + |c1 - c2| + |s2 - X|
cross aisles X: LC = 4.8 m, CC = 29.5 m, RC = 50.2 m (RC only for walkways W08-W17)
```

The I/O depot is the green box on the case drawing, which sits on the central corridor at support point CC-08 (29.5 m along, 20.3 m across). All 2,339 locations (2,292 from the file plus 47 imputed) were placed in this network, and the distance from the I/O to every location was pre-computed.

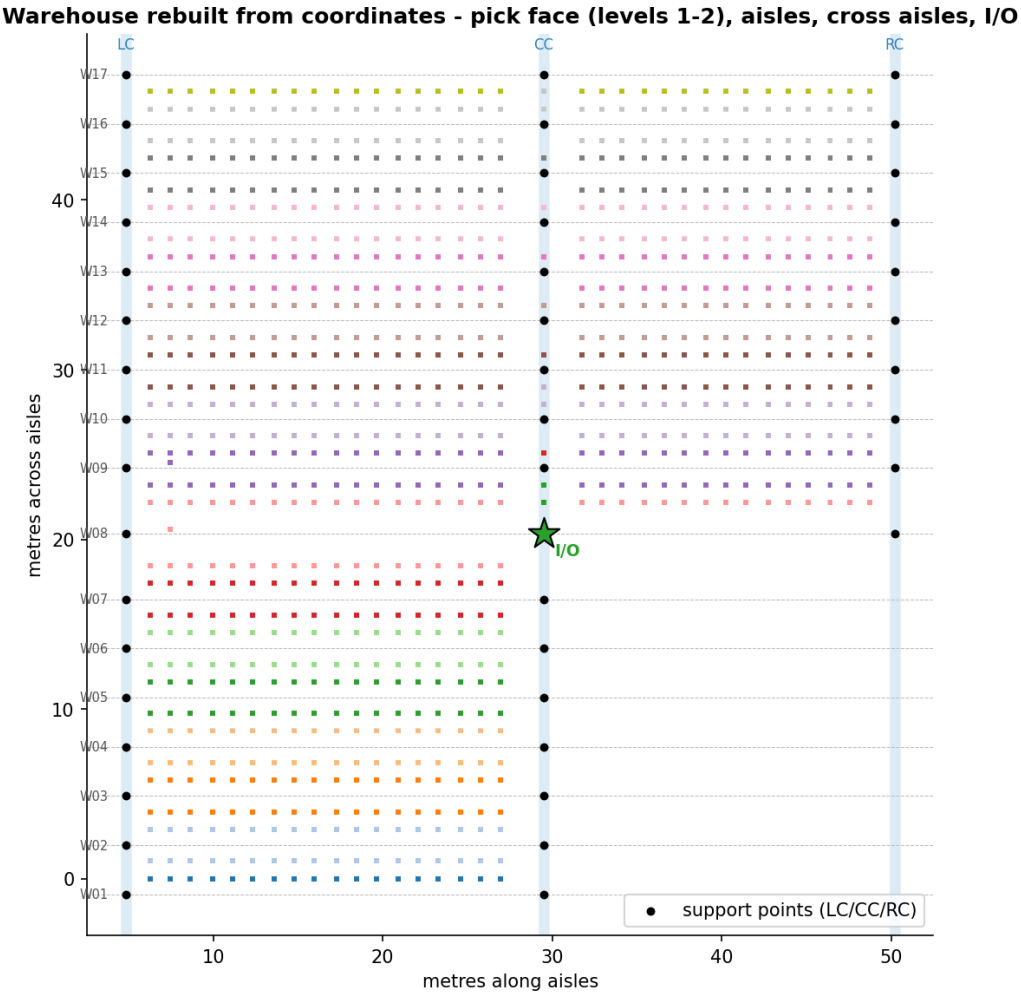


Figure 4. The warehouse rebuilt from the coordinates (levels 1-2, one colour per walkway), with the 17 walkway centre-lines (W01-W17), the three cross aisles (LC, CC, RC), the support points and the I/O.

What it shows. Each coloured square is a hand-pick position, coloured by the walkway it opens onto. Dashed horizontal lines are walkway centre-lines, the blue vertical bands are the left, central and right cross aisles, and the black dots are the 44 support points supplied in the case. The lower block (W01-W07) exists only to the left of the central corridor; the upper block (W08-W17) extends on both sides.

How the code produced it. build_geometry() detected that x runs along the walkways and y across them, assigned faces to walkways using the rack-partner rule, calibrated metres per axis, and draw_frame() plotted the result.

What it means. The rebuilt network reproduces the CAD layout of Figure 1, including the wide walkway W08 at the gap between the two blocks, where the I/O sits. This check matters: every distance in the report is computed on this network.

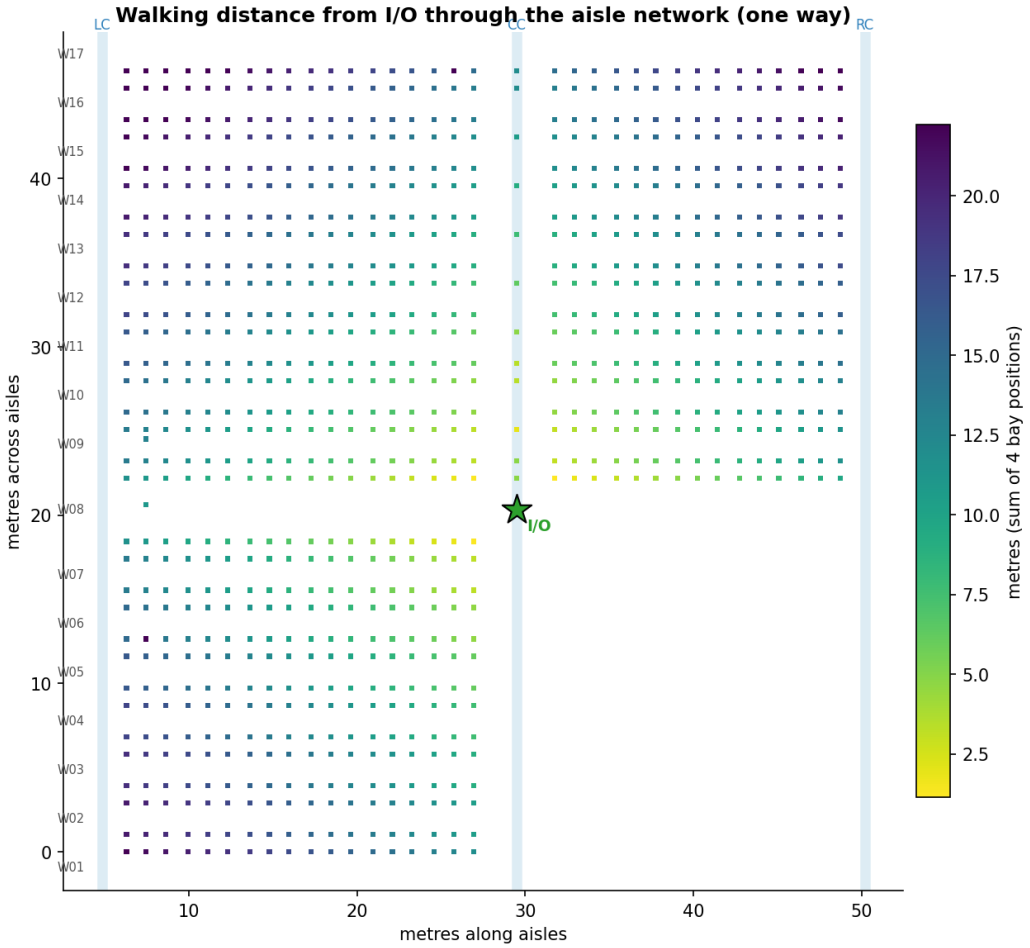


Figure 5. *One-way walking distance from the I/O to every pick position through the aisle network.*

What it shows. The colour gives the walking distance from the I/O to each position (yellow = close, purple = far). The closest positions are about 1.5-2.5 m away, on walkways W08-W10 next to the central corridor; the farthest corners of the building are about 22 m away.

How the code produced it. Network.one_to_many() computed the network distance from the I/O node to all 2,339 locations; heat_layer() coloured them.

What it means. This map is the 'value' of every shelf. The slotting optimisation ranks shelves by this distance and gives the best ones to the most frequently picked items. It also shows that distance is not symmetric around the I/O: positions just above it (W08-W10) are cheaper to reach than positions in the lower block because the I/O sits on walkway W08.

6. Exploratory data analysis

The EDA describes demand, orders, waves, products and operators, and looks for patterns that the optimisation can exploit. The period covered is **5 January to 19 October 2023**: 32,634 orders from 588 customers on 178 days with orders, an average of 183 orders per day.

6.1 Order intake

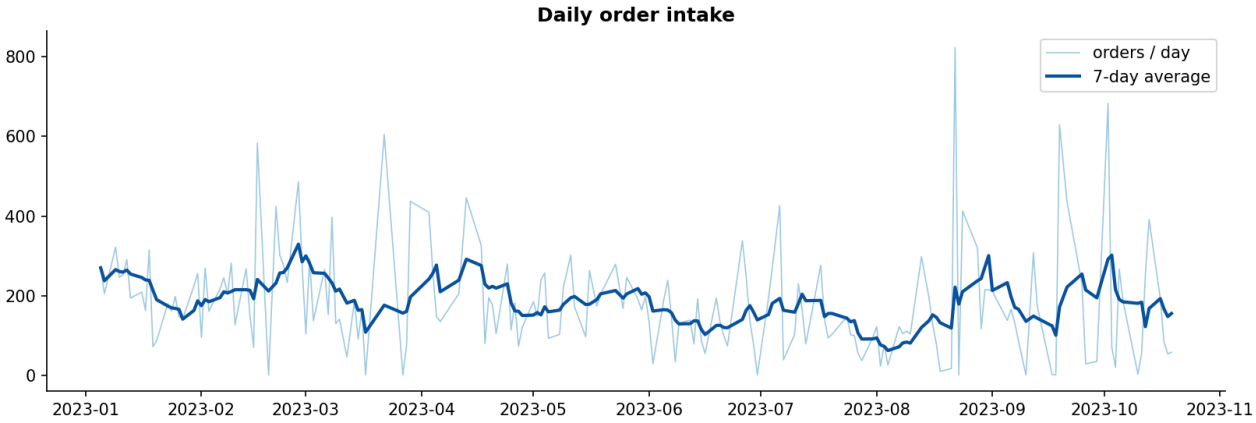


Figure 6. *Daily number of orders (thin line) and 7-day moving average (thick line).*

What it shows. Daily order counts vary strongly, from zero to more than 800, around a 7-day average that moves between about 100 and 330 orders per day. Intake dips in late July and early August and shows isolated spikes in late August, September and early October.

How the code produced it. The code counts distinct order numbers per creation date in Customer_Order.csv and adds a rolling 7-day mean (section 5 of main()).

What it means. Demand is lumpy, so the warehouse must cope with peaks several times the average. The summer dip is consistent with the Quebec construction holiday (late July), plausible for a safety-boot supplier to construction customers. A layout that makes every wave shorter helps most on spike days, when picking capacity is the bottleneck.

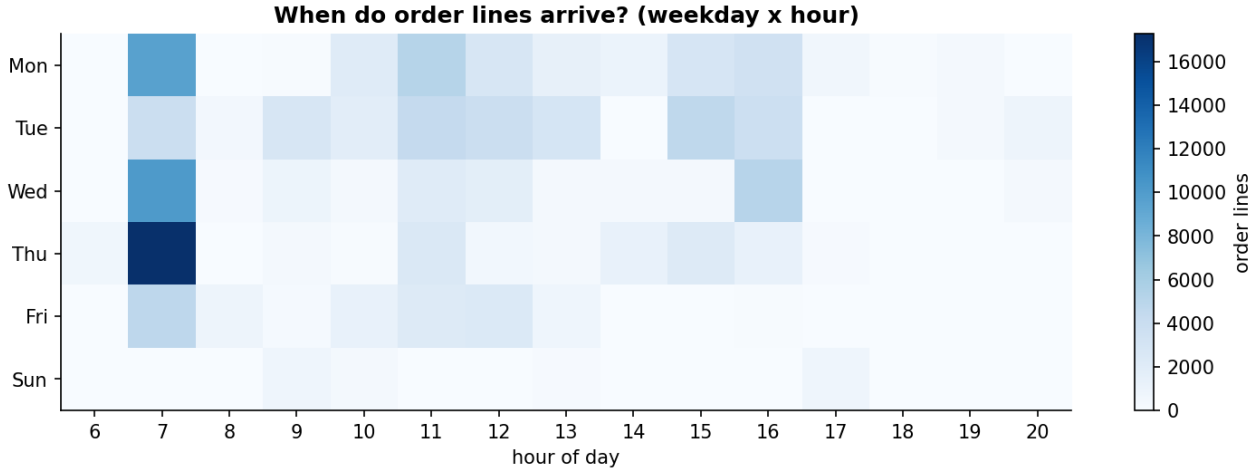


Figure 7. *Order lines by weekday and hour of creation.*

What it shows. Most order lines arrive in one burst at 7 am, strongest on Thursday (about 17,000 lines over the period), then Monday and Wednesday. Smaller waves of lines arrive around 11 am and 3-4 pm. Saturday has no orders and Sunday almost none.

How the code produced it. Order lines grouped by the day of week and the hour of creationDate.

What it means. Orders are released in batches (for example an overnight EDI load at 7 am) rather than trickling in. This is an opportunity for **order batching**: when hundreds of lines are known at 7 am, waves could be built by location proximity instead of arrival order. It also explains why the measured waiting time between order creation and wave release is zero (Section 11).

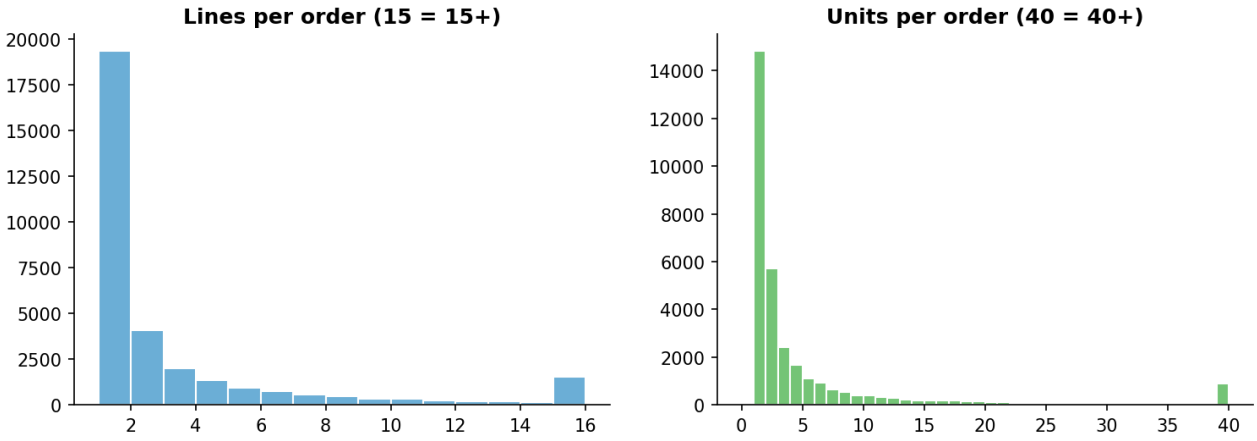


Figure 8. Distribution of lines per order (left) and units per order (right); the last bar collects all larger values.

What it shows. About 19,000 of the 32,634 orders (59%) have a single line and about 14,800 contain a single unit. The mean is 3.75 lines and 6.99 units per order, pulled up by a long tail of large orders (up to 408 lines and 2,400 units).

How the code produced it. Order lines grouped by order number in Customer_Order.csv.

What it means. Most orders are small, so a wave of 24 units typically combines several orders (mean 4.1 orders per wave). Since a single order cannot fill a trolley, the travel per unit depends heavily on where the items of *different* orders are stored, which is exactly what slotting controls.

6.2 Waves

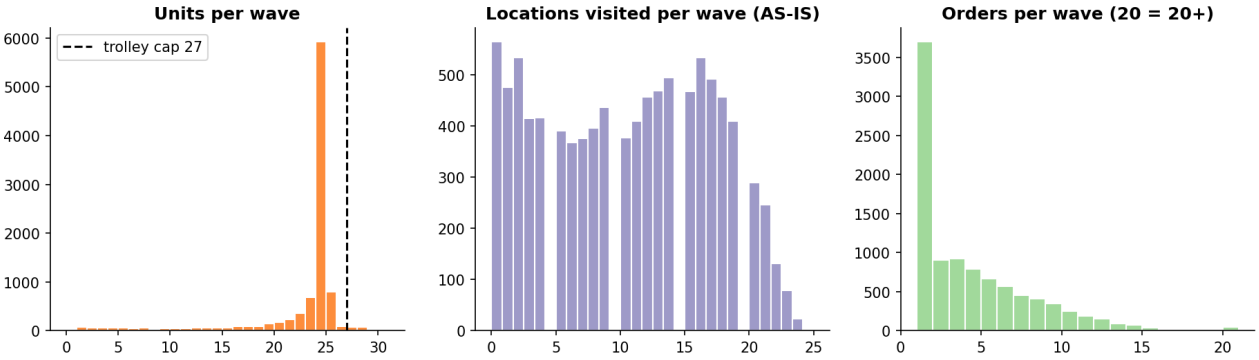


Figure 9. Units per wave (left, dashed line = trolley capacity of 27), locations visited per wave under the current layout (centre) and orders per wave (right).

What it shows. Most waves contain 24 units (about 5,900 waves), so the WMS fills the trolley to roughly 24 of 27 places; 74 waves exceed the 27-item capacity. A wave visits on average 10.4 distinct locations today (0 to 25). About 3,700 waves (38%) serve a single order. The regular gaps in the centre histogram are a binning artefact of the chart, not a feature of the data.

How the code produced it. Picking_Wave.csv grouped by wave; 'locations visited' counts distinct historical locations per wave; orders per wave comes from Customer_Order.csv.

What it means. With about 10 stops per wave, each wave is a small travelling-salesman problem, so the order of the stops matters (Section 7.2). The 74 over-capacity waves should be checked with the company, since they break a physical constraint.

6.3 Product demand and ABC

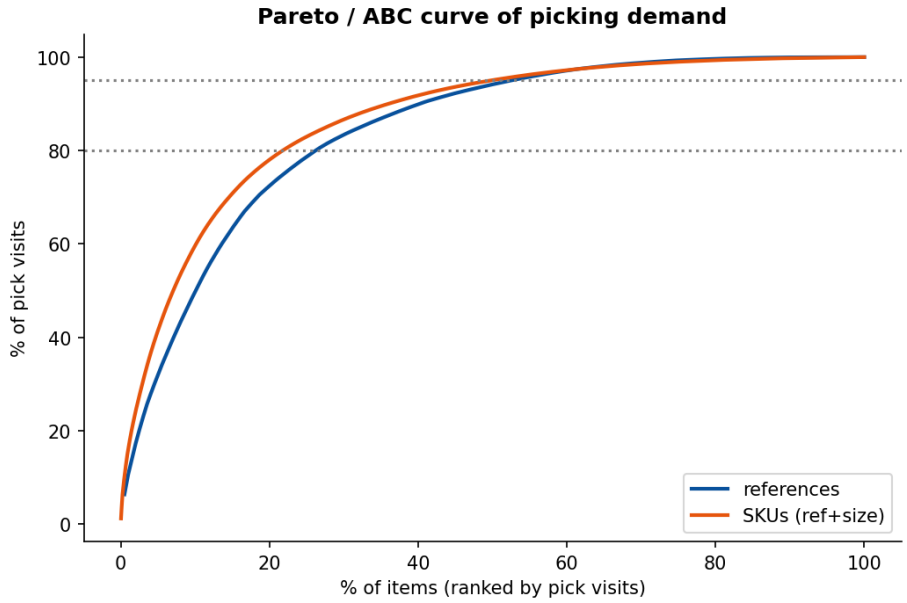


Figure 10. *Pareto curve of picking demand for references (blue) and SKUs = reference + size (orange). Dotted lines mark 80% and 95% of pick visits.*

What it shows. About 20% of SKUs account for roughly 78% of all pick visits, and about 22% of SKUs reach the 80% line. At SKU level the classes are: **A = 536 SKUs (80% of visits), B = 668 SKUs (15%), C = 1,080 SKUs (5%)**. At reference level: A = 53, B = 50, C = 87 references, and 8 references had no demand in the design period.

How the code produced it. build_demand() counts **pick visits** per item, i.e. the number of distinct waves in which the item appears, sorts items, accumulates their share and cuts at 80% and 95% (abc_from_counts()).

What it means. Demand is highly concentrated, which is the precondition for slotting to work: if a small set of items makes most trips, putting those items close to the I/O shortens most trips. Visits rather than units were used because walking happens per visit: six pairs taken from the same shelf cost one stop, not six.

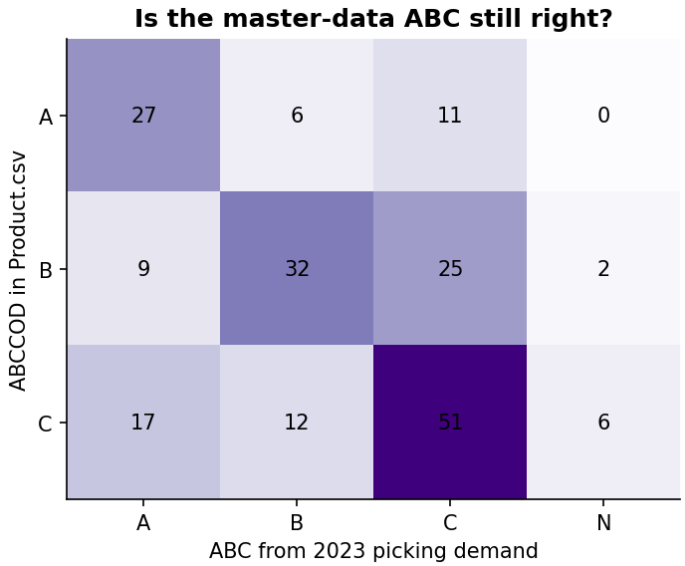


Figure 11. *Cross-tabulation of the ABC class recorded in Product.csv (rows) against the ABC class computed from 2023 picking demand (columns). N = no demand in the design period.*

What it shows. Only 110 of 198 picked references (56%) sit on the diagonal. Seventeen references recorded as C are actually class A, and eleven recorded as A are actually C. The most-picked reference of the year, 8N10W9 (18,069 units), is recorded as B, and the fourth, SMMRK3, as C.

How the code produced it. The demand-based ABC (Figure 10) joined to the ABCCOD column of Product.csv with pandas.crosstab.

What it means. The master data is out of date. Any class-based storage that relies on the recorded ABCCOD puts many fast movers in the wrong zone. This is one reason why the case's Class-Based policy performs well below the proposed design, and it leads to the recommendation to refresh the classes from WMS data regularly.

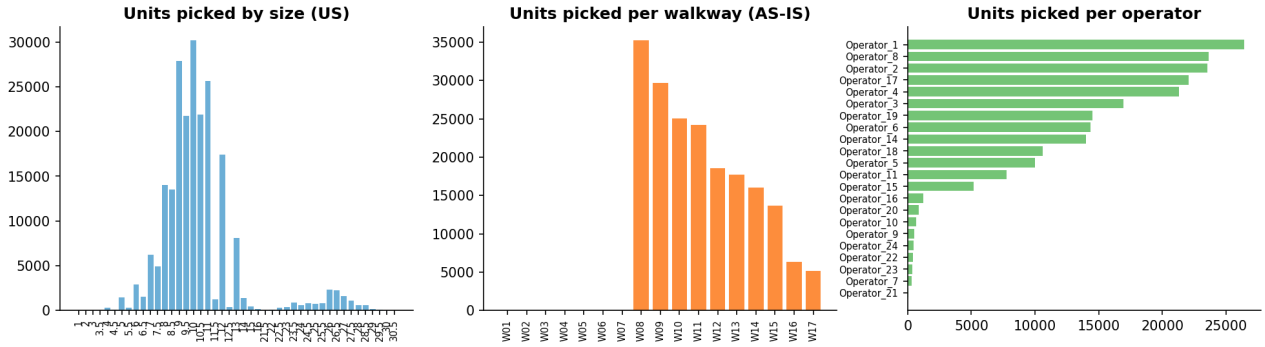


Figure 12. *Units picked by shoe size (left), by walkway under the current operation (centre) and by operator (right).*

What it shows. Sizes: US 8 to 11 dominate. A second group of sizes between 21.5 and 30.5 also appears, which looks like a different sizing system (for example Mondopoint or European) and is worth confirming. **Walkways:** picking today is concentrated on walkway W08 (35,000 units), the walkway that passes the I/O, and falls off steadily to W17 (5,000 units); **walkways W01-W07, the entire lower block, recorded zero picks**. **Operators:** five operators each picked more than 21,000 units, while nine picked fewer than 1,300.

How the code produced it. Picking_Wave joined to the location table (for the walkway) and grouped by size, walkway and operator.

What it means. The current operation already concentrates picking in the upper block near the I/O, so the lower block holds stock that is never picked from. The operator chart anticipates the workload imbalance analysed in Section 10.

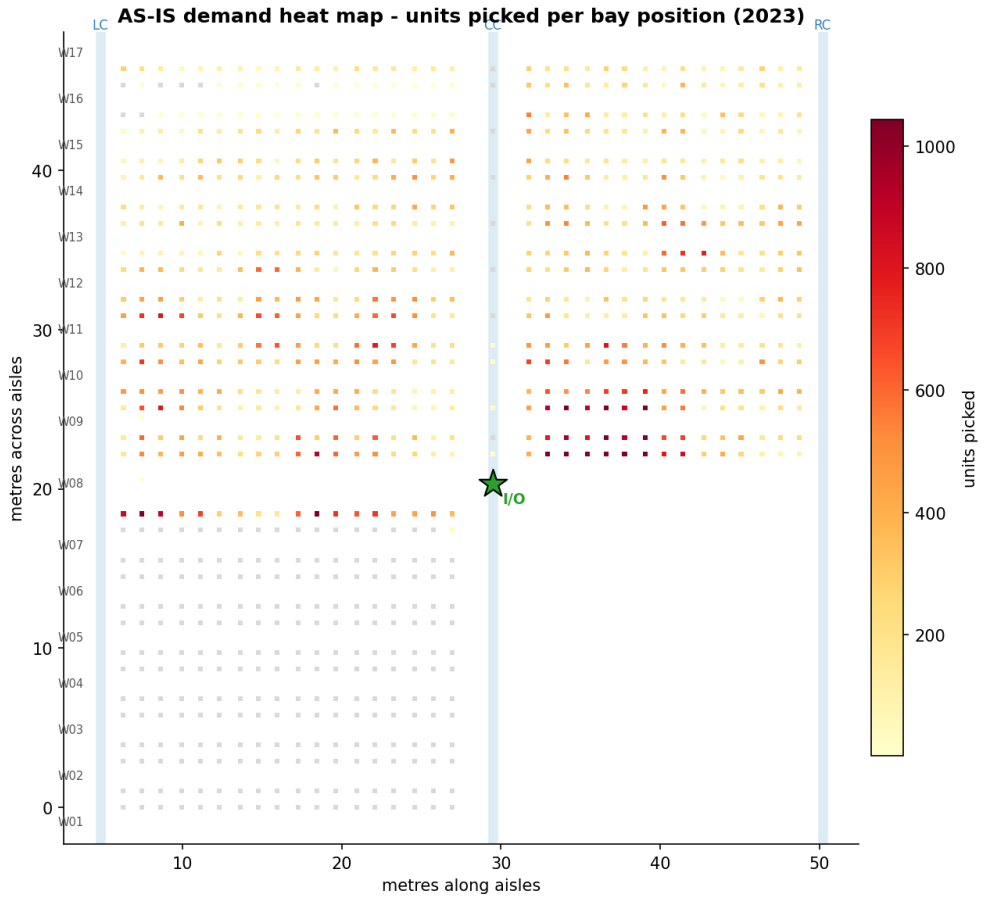


Figure 13. *Demand heat map of the current operation: units picked per bay position in 2023 (grey = no picks).*

What it shows. Picking is spread over the whole upper block, with hot spots on the right block of walkways W08-W10 and on the row just below the I/O (faces I-23 on walkway W08). The lower block is grey.

How the code produced it. Each historical pick placed on its recorded location; units summed per plan position over levels 1-2 (heat_layer()).

What it means. Even though picking happens in the upper half, hot items are scattered across it rather than packed next to the I/O. The mean one-way distance from the I/O to a pick is 21.0 m, and only 30% of picks are within 15 m of the I/O. This is the inefficiency the new slotting removes.

7. Optimisation methods

7.1 How each storage arrangement is evaluated

The four case policy files list, for every location, which 18 boxes it holds. A product usually has many boxes spread over many locations, so for each policy the code must decide which box a picker would take. The same rule is used for every arrangement to keep the comparison fair: **take the item from the nearest hand-pick location (levels 1-2) that holds it**, measured by network distance from the I/O; use levels 3-4 only if the item is not on the pick face; if the exact size is not stocked, use the nearest box of the same reference (assign_pick_locations()). AS-IS is different: it uses the location actually recorded in Picking_Wave.csv.

7.2 Routing: solving a small travelling-salesman problem for every wave

Each wave starts and ends at the I/O and must visit its set of locations (on average about 10). Finding the shortest such tour is a travelling-salesman problem (TSP) on the aisle network. Three routing methods were compared:

Method	How it works	Role in the study
Recorded	The order in which the WMS listed the picks in Picking_Wave.csv; consecutive picks at the same location are merged into one stop.	Today's baseline (AS-IS only).
Serpentine	Visit walkways in order, alternating direction in each (S-shape style); both sweep directions are tried and the shorter one kept.	A simple rule a printed pick list could follow.
Nearest-neighbour + 2-opt	Start at the I/O, always go to the closest unvisited stop (nearest neighbour); then repeatedly reverse any segment of the tour whose reversal makes it shorter (2-opt, Croes 1958) until no improvement remains.	The optimised route used for all main results.

Table 6. *Routing methods (tour_nn_2opt(), tour_serpentine()).*

For each wave the code builds the matrix of network distances between the I/O and the stops (Network.matrix()), runs the routing methods and stores the tour length. Identical stop sets are cached so each route is solved only once. In total about 145,000 wave routes were evaluated.

7.3 Slotting: designing the new arrangement

The new arrangement uses the **Cube-per-Order Index (COI)** principle (Heskett, 1963), one of the classic rules of warehouse design: items that are picked most often per unit of space they occupy should get the locations closest to the depot. The design_slotting() function applies it in four steps.

- **Size the forward zone per item.** Each item with demand gets enough pick-face slots for two days of its average daily demand, q = max(1, ceil(2 x daily units)), measured on the training period (120 working days).
- **Rank items** by COI density = pick visits / q. An item visited often relative to the space it needs ranks first.
- **Rank shelves** by network distance from the I/O (Figure 5), hand-pick levels only.
- **Assign** the best-ranked items to the best-ranked shelves, slot by slot, until every item has its forward stock. The result is a compact forward zone of 3,372 slots for 1,769 items.

Two variants were built. **COI** ranks every SKU individually. **FAMILY** ranks boot models (references) by their combined density and keeps all sizes of the same model next to each other, because one order often contains several sizes of the same model and they can then be picked at one stop. The variant to recommend was chosen on training waves only: **FAMILY won (65.9 m vs 71.2 m per wave)**.

From the ideal design to a practical plan: the minimum-move swap

Rebuilding the whole warehouse from scratch would mean moving almost all 41,256 boxes. min_move_layout() re-slots **only the forward-zone locations**, i.e. those nearest the I/O. For each forward-zone slot it keeps boxes that are already correct, otherwise brings the required box from its nearest current position and puts the displaced box into the slot just vacated (a swap). All other locations keep today's stock. Because the forward zone is by construction the closest part of the rack, walking performance equals that of the ideal design. The resulting plan moves 6,738 units (16.3% of stock) into 186 locations.

7.4 From distance to time

Picking_Wave.csv has no timestamps, so time is estimated with a standard engineering time model. The parameters are conventional values and are tested in the sensitivity analysis (Section 11.3).

Component	Formula	Value used
Walking	distance / walking speed	1.0 m/s with a trolley
Stopping and searching	number of stops x time per stop	10 s per stop
Picking	units x time per unit	6 s per unit
Setup	once per wave (take list and trolley, drop at staging)	60 s

Table 7. *Time model per wave.*

Productivity in units per hour is units divided by this estimated picking time. It excludes breaks and idle time, so it is a measure of picking efficiency rather than of shift output.

7.5 Validation design

- **Out-of-sample test.** Waves were sorted by release time. The new layout was designed using demand from the first 70% (6,823 waves up to 17 July 2023) and tested separately on the last 30% (2,549 comparable waves).
- **Paired statistical test.** Because every arrangement processes the same waves, the Wilcoxon signed-rank test (Wilcoxon, 1945) compares wave by wave, without assuming normal distributions.
- **Sensitivity analysis.** The time saving was recomputed for walking speeds of 0.8, 1.0 and 1.2 m/s and pick times of 4, 6 and 10 s per unit.
- **Decomposition.** AS-IS is compared with AS-IS re-routed (same locations, optimised route) and then with the new layout (optimised route), which separates the effect of routing from the effect of slotting.

8. Results: storage arrangement and routing

8.1 Where picking happens under each arrangement

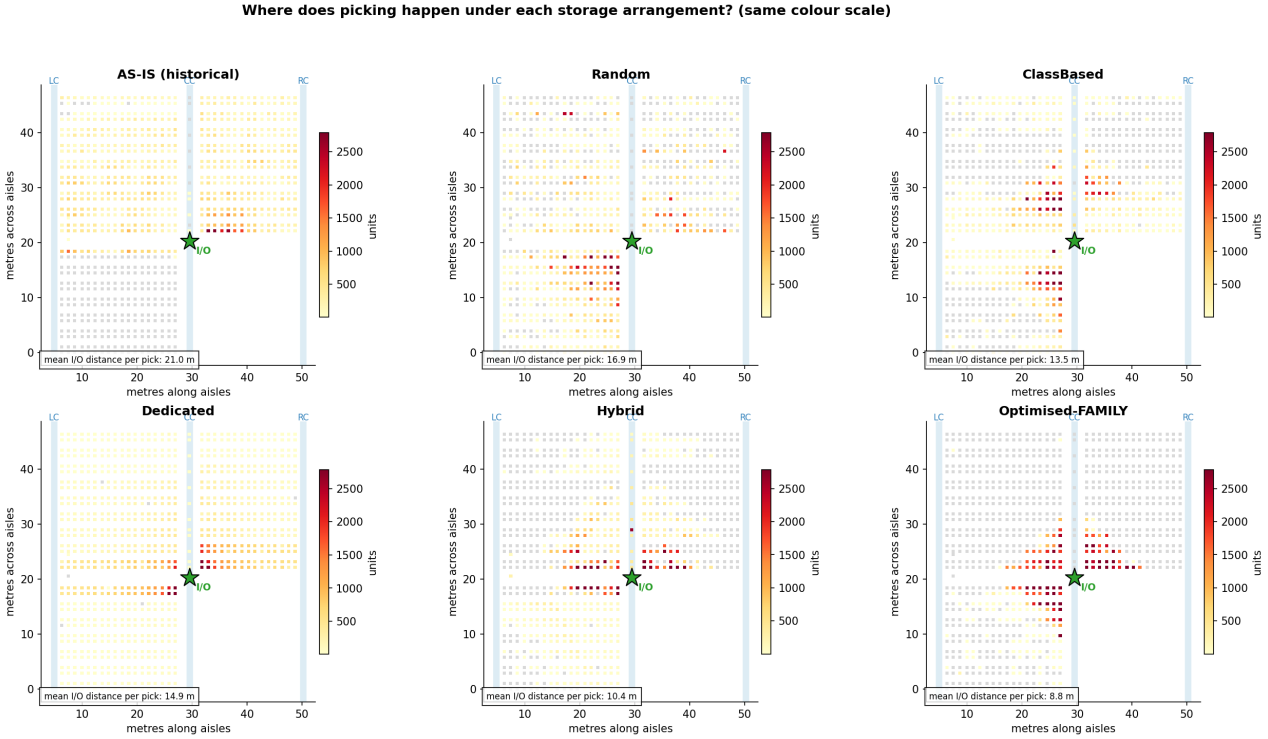


Figure 14. *Units picked per bay position under each arrangement, on the same colour scale. The box in each panel gives the mean one-way distance from the I/O to a pick.*

What it shows. AS-IS spreads picks thinly over the upper block (mean 21.0 m from the I/O). Random storage, applied to the case stock file, puts the nearest copy of many items in the lower block close to the central corridor (16.9 m). Class-Based concentrates near the central corridor (13.5 m), Dedicated spreads each SKU's single home along walkways W07-W09 (14.9 m), Hybrid clusters fast movers near the I/O (10.4 m), and **Optimised-FAMILY packs picking into a tight zone around the I/O (8.8 m)**.

How the code produced it. For each arrangement, assign_pick_locations() placed every pick, and heat_layer() summed units per position.

What it means. The mean pick distance falls by 58% from AS-IS to the optimised layout, and 99% of optimised picks are within 15 m of the I/O, compared with 30% today. The number of distinct pick locations used falls from 1,205 (AS-IS) and 1,954 (Dedicated) to 281, so pickers work in a much smaller area.

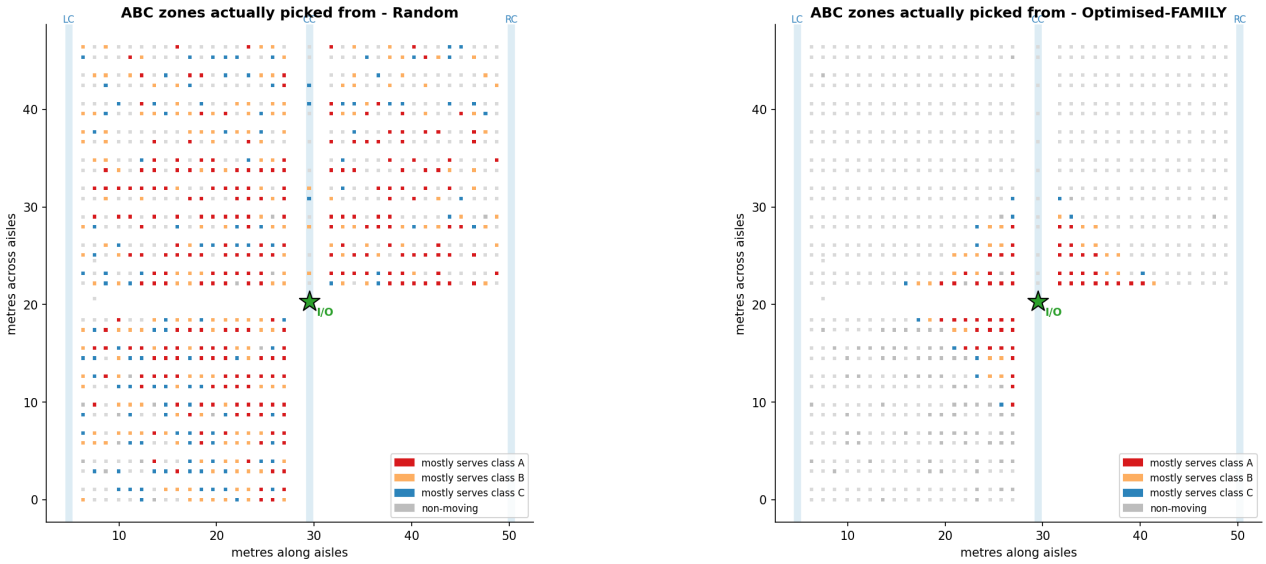


Figure 15. ABC class of the demand each pick position mostly serves: case Random storage (left) vs the proposed Optimised-FAMILY layout (right).

What it shows. Under Random storage, class-A (red), B (orange) and C (blue) positions are mixed everywhere. Under the optimised layout they form rings: class A immediately around the I/O on both sides of the central corridor, class B outside it, class C at the edge of the forward zone, and the rest of the rack (grey) holds stock that is not picked from.

How the code produced it. For every position, the class of the units actually picked from it was computed and the dominant class plotted.

What it means. This is the visual signature of a COI/ABC slotting: travel is short because the items that generate most trips are where trips are shortest.

8.2 Walking distance per wave

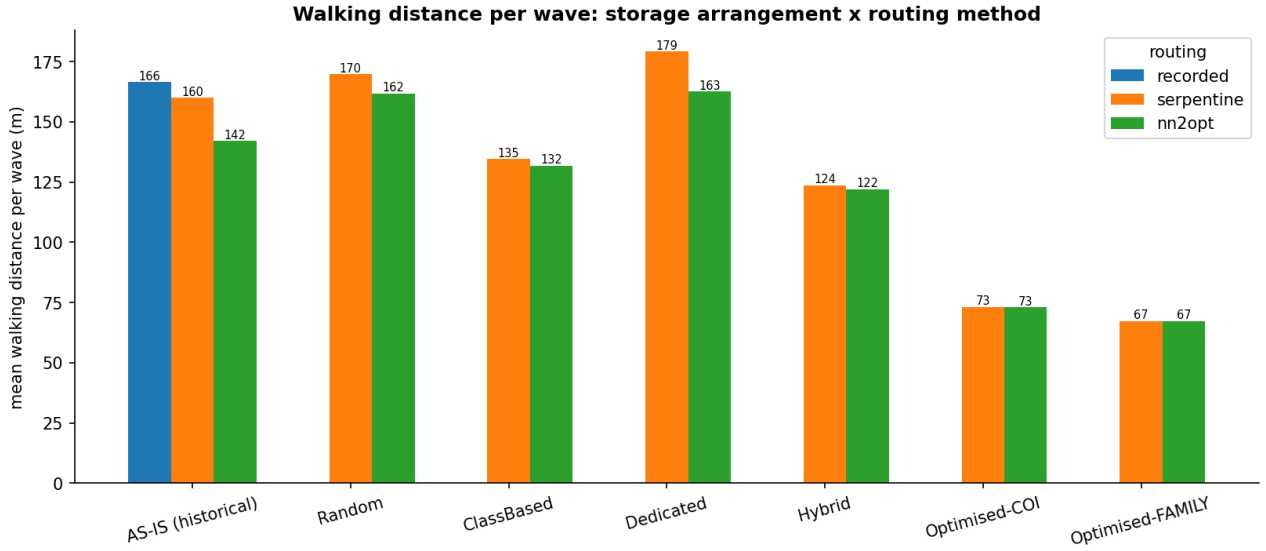


Figure 16. Mean walking distance per wave for each arrangement and routing method (7,341 comparable waves).

What it shows. AS-IS walks 166 m per wave with the recorded sequence, 160 m with the serpentine rule and 142 m with the optimised route. Among the case policies with optimised routing: Hybrid 122 m, Class-Based 132 m, Random 162 m, Dedicated 163 m. The proposed layouts reach 73 m (COI) and **67 m (FAMILY)**.

How the code produced it. evaluate_scenarios() routed every wave under every combination; scenario_summary() averaged the results.

What it means. Three conclusions. (1) Routing matters when stops are spread out: for AS-IS and Dedicated, the optimised route saves 18 m and 16 m per wave over the simple serpentine rule, and serpentine Dedicated (179 m) is worse than today. (2) Among the case policies, those that put fast movers near the I/O (Hybrid, Class-Based) work; Random and Dedicated barely beat AS-IS. Dedicated suffers because each SKU is tied to fixed positions spread across the rack (1,954 different locations are used). (3) Once fast movers are packed around the I/O, the routing method no longer matters (67 m either way), so the proposed layout is robust even if a simple pick-list order is used.

Arrangement	Routing	Mean m	Median m	P90 m	Stops	Walkways	Min/wave	Units/h	vs AS-IS
AS-IS	recorded	166.4	168.0	271.8	11.2	4.55	7.85	168	-
AS-IS	serpentine	159.9	161.5	260.1	11.2	4.55	7.73	171	-3.9%
AS-IS	NN + 2-opt	142.1	146.0	220.4	11.2	4.55	7.44	178	-14.6%
Random	NN + 2-opt	161.7	173.1	257.5	10.9	5.92	7.71	172	-2.8%
Class-Based	NN + 2-opt	131.7	137.1	210.9	10.8	5.39	7.20	184	-20.8%
Dedicated	NN + 2-opt	162.6	171.8	266.3	11.2	5.52	7.78	170	-2.3%
Hybrid	NN + 2-opt	122.0	124.6	211.5	10.7	4.42	7.02	188	-26.7%
Optimised-COI	NN + 2-opt	73.1	77.0	108.0	10.3	3.50	6.14	215	-56.1%
Optimised-FAMILY	NN + 2-opt	**67.3**	**69.7**	**104.5**	**8.7**	**3.28**	**5.78**	**229**	**-59.5%**

Table 8. *Scenario summary on the 7,341 waves fully placed in every scenario (01_Analysis_Tables.xlsx, Scenario_Summary).*

FAMILY also needs fewer **stops** per wave (8.7 vs 11.2), because sizes of the same model now share locations, and fewer **walkways** are entered (3.3 vs 4.6). Both reduce time beyond distance alone.

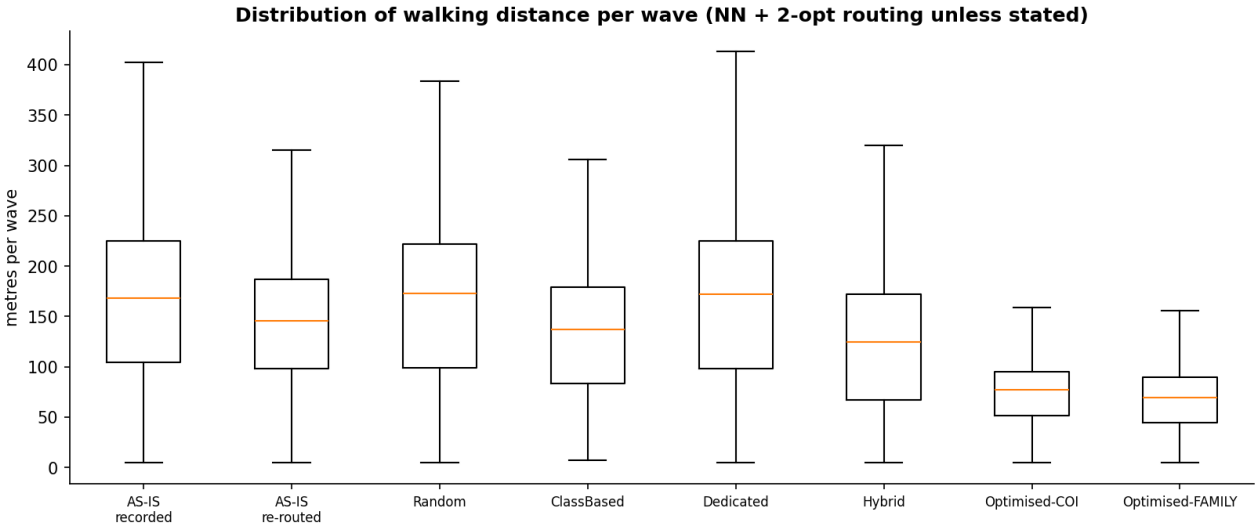


Figure 17. *Distribution of walking distance per wave (boxes = middle 50% of waves, line = median, whiskers = typical range).*

What it shows. The optimised layouts have both a lower and a much **narrower** distribution: the middle 50% of Optimised-FAMILY waves walk 45-90 m, against 104-225 m for AS-IS, and the longest typical waves fall from about 400 m to about 155 m.

How the code produced it. Box plots of the per-wave distances produced by evaluate_scenarios().

What it means. Less variability means more predictable wave times, which makes it easier to plan staff and promise dispatch times. The long, expensive waves are the ones that disappear.

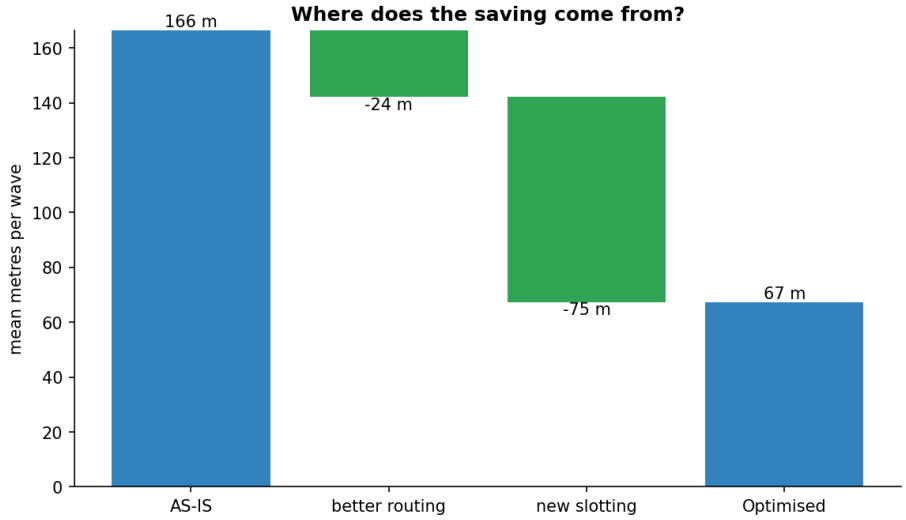


Figure 18. Decomposition of the saving per wave: AS-IS, effect of better routing, effect of new slotting, optimised result.

What it shows. Starting from 166 m, better routing saves 24 m (14.6%) and the new slotting saves a further 75 m (44.9 percentage points), ending at 67 m.

How the code produced it. The difference between AS-IS recorded and AS-IS re-routed isolates routing; the difference between AS-IS re-routed and Optimised re-routed isolates slotting (Saving_Decomposition sheet).

What it means. Three-quarters of the gain comes from slotting, but the routing part needs no physical change at all and can be implemented first.

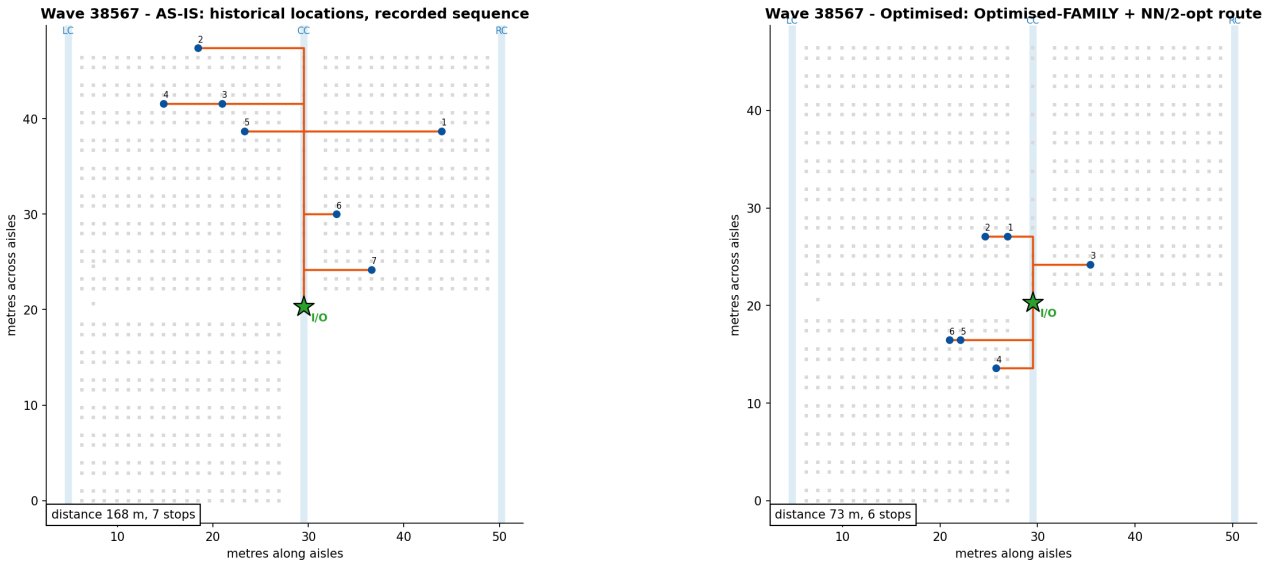


Figure 19. The same real wave (38567) under the current operation (left) and the proposed layout with optimised routing (right). Numbers give the visiting order.

What it shows. Today this wave crosses the building: 7 stops, up to walkway W17 and both ends of the rack, 168 m. With the new layout and route, the same items are found in 6 stops within a few metres of the I/O, for 73 m.

How the code produced it. routes stored for selected scenarios by evaluate_scenarios(); Network.path() draws each leg along the aisles and cross aisles.

What it means. A concrete illustration of both levers: the items move closer (slotting) and the walk visits them in a sensible order (routing). Every wave's stop list, before and after, is in 03_Reconstructed_Picking_Routes.xlsx.

8.3 Are the results statistically robust?

Comparison (paired, 7,341 waves)	Mean A	Mean B	Median saving	Waves improved	p-value
Routing only: AS-IS recorded vs AS-IS re-routed	166.4 m	142.1 m	16.0 m	73.3%	< 0.001
Slotting only: AS-IS re-routed vs Optimised	142.1 m	67.3 m	75.1 m	97.2%	< 0.001
Both: AS-IS recorded vs Optimised	166.4 m	67.3 m	95.6 m	97.5%	< 0.001
Random vs Optimised	161.7 m	67.3 m	98.0 m	94.8%	< 0.001
Class-Based vs Optimised	131.7 m	67.3 m	62.1 m	94.0%	< 0.001
Dedicated vs Optimised	162.6 m	67.3 m	97.4 m	92.9%	< 0.001
Hybrid vs Optimised	122.0 m	67.3 m	50.1 m	84.8%	< 0.001

Table 9. *Wilcoxon signed-rank tests (Statistical_Tests sheet). The p-values are effectively zero.*

Test waves only (2,549, after 17 July 2023)	Mean distance	Saving vs AS-IS
AS-IS recorded	170.1 m	-
AS-IS re-routed	146.4 m	-13.9%
Hybrid (best case policy)	125.4 m	-26.3%
Optimised-COI	76.7 m	-54.9%
Optimised-FAMILY	**70.0 m**	**-58.8%**

Table 10. *Out-of-sample result: the layout was designed without these waves.*

The saving on waves the design never saw (-58.8%) is almost identical to the full-year saving (-59.5%), so the layout is not tuned to past noise: demand concentration is stable enough over the year for a forward zone to keep working. Every individual comparison is significant, and even against the best case policy (Hybrid) the proposed layout wins on 85% of waves.

9. Results: picking time and its variation

9.1 Where the picker's time goes

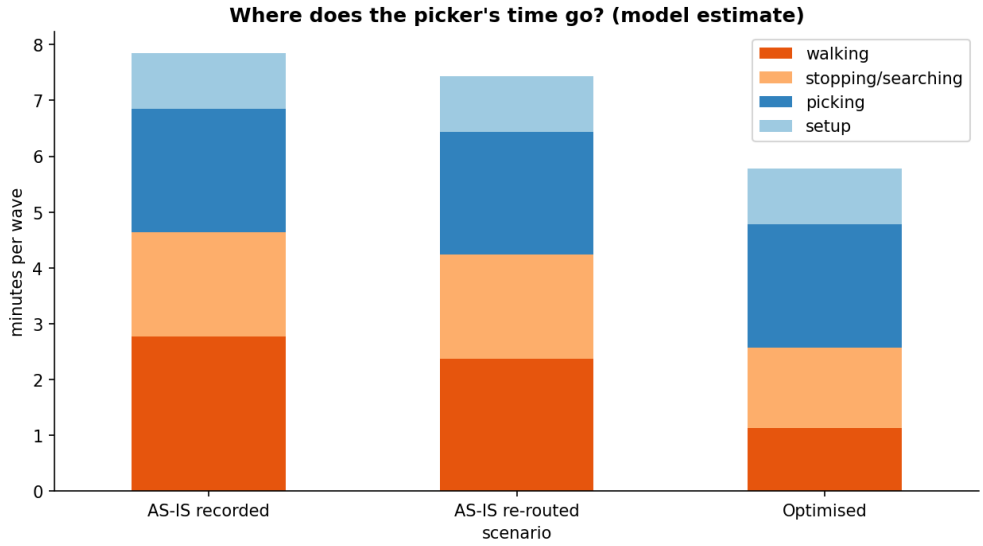


Figure 20. *Estimated time per wave split into walking, stopping/searching, picking and setup for AS-IS, AS-IS re-routed and the optimised layout.*

What it shows. An AS-IS wave takes 7.8 min: 2.8 min walking, 1.9 min stopping, 2.2 min picking and 1.0 min setup. Re-routing trims walking to 2.4 min. The optimised layout cuts walking to 1.1 min and stopping to 1.5 min (fewer stops), for 5.8 min in total.

How the code produced it. The time model of Table 7 applied to each wave's distance, stops and units.

What it means. Walking is 35% of picker time today and 19% after optimisation. Picking and setup (3.2 min per wave) are untouched by layout and routing, which is why a 59.5% distance saving becomes a 26.3% time saving. Further gains would have to come from handling time or wave building.

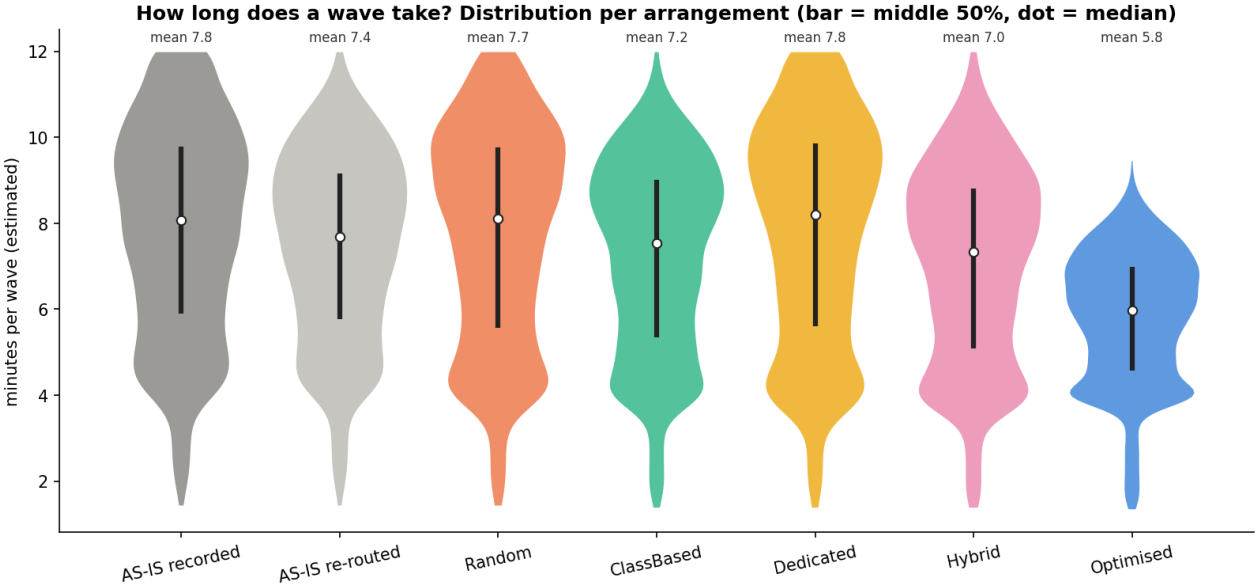


Figure 21. *Distribution of minutes per wave for every arrangement (violin = distribution, bar = middle 50%, dot = median).*

What it shows. Means are AS-IS 7.8, re-routed 7.4, Random 7.7, Class-Based 7.2, Dedicated 7.8, Hybrid 7.0 and Optimised 5.8 min. Every distribution except Optimised has two peaks, short waves

around 4.5 min and long ones around 9-10 min. Optimised removes most of the long peak (90th percentile 7.7 min vs 11.0 min today).

How the code produced it. time_variation() combined all arrangements and drew one violin per arrangement.

What it means. The long waves are those whose items are scattered, and they are exactly what slotting removes. The coefficient of variation falls from 0.32 to 0.27, so wave times also become more predictable.

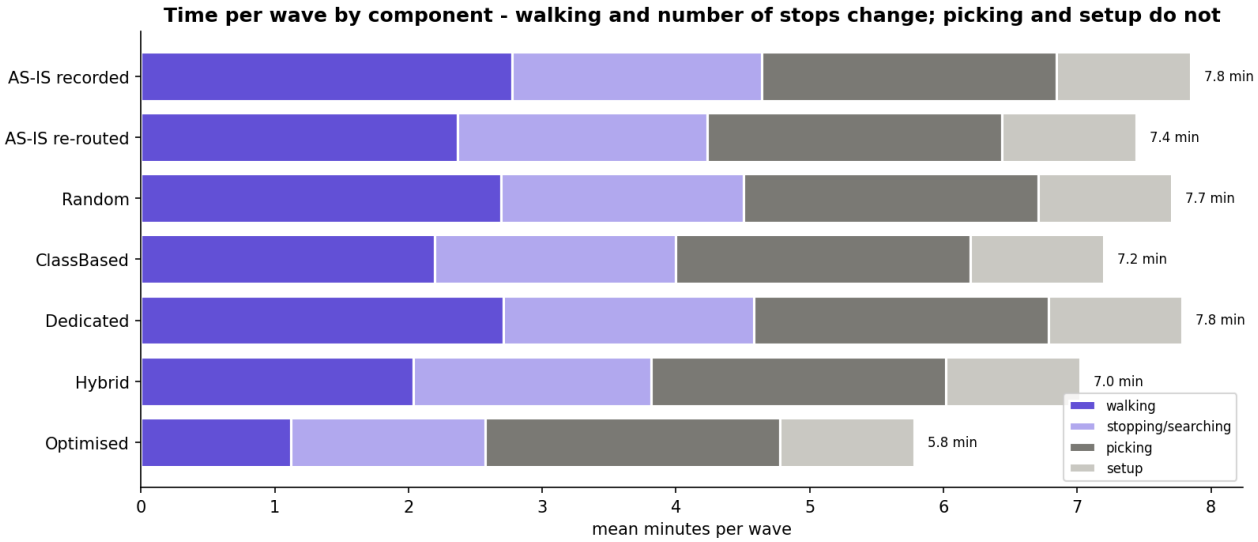


Figure 22. Mean time per wave by component for every arrangement.

What it shows. Picking (2.2 min) and setup (1.0 min) are the same in every arrangement because the waves are the same. The differences are in walking (2.0-2.8 min for the case policies, 1.1 min optimised) and stopping (1.8-1.9 min vs 1.5 min).

How the code produced it. Mean of each time component per arrangement (Time_Variation_Stats sheet).

What it means. This confirms where the saving comes from and shows that the comparison is fair: arrangements differ only in the parts of the work that the arrangement can influence.

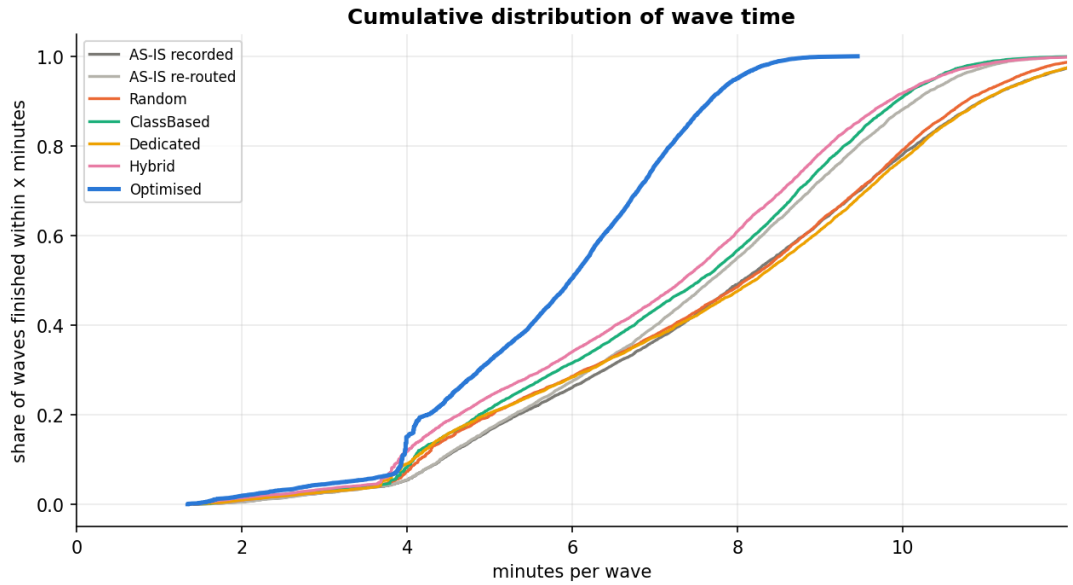


Figure 23. Cumulative share of waves completed within x minutes, per arrangement.

What it shows. With the optimised layout about 80% of waves finish within about 7 minutes; with AS-IS 80% finish within about 10 minutes. The step near 4 minutes in every curve comes from single-location waves (one item, about 24 units), whose time is almost entirely picking and setup.

How the code produced it. Sorted wave times per arrangement plotted against their cumulative share.

What it means. The optimised curve lies to the left of all others everywhere, meaning that it is faster not only on average but at every service level. This is the right view for promising dispatch times, e.g. '95% of waves done within 8 minutes'.

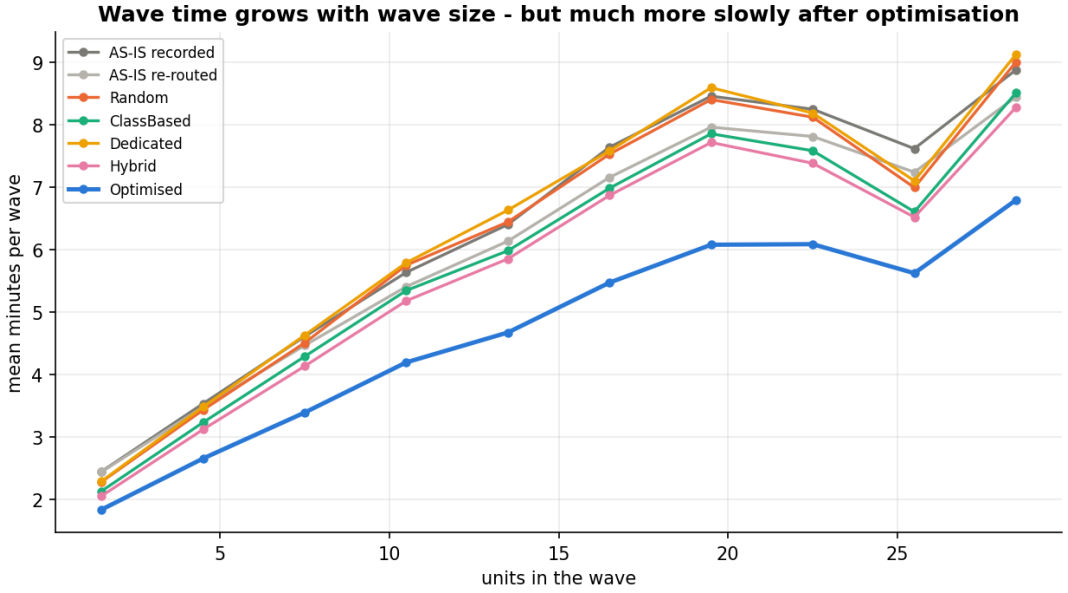


Figure 24. *Mean minutes per wave by number of units in the wave.*

What it shows. Wave time rises with wave size in every arrangement, but more slowly in the optimised layout: a 20-unit wave takes about 8.5 min today and 6.1 min optimised. The dip at 25-26 units reflects full-trolley waves that are dominated by one or two items.

How the code produced it. Waves grouped into 3-unit bins of size; mean time per bin (Time_by_WaveSize sheet).

What it means. Each additional item costs less walking in the optimised layout because it is probably near the I/O. This makes larger waves (fuller trolleys or batched orders) more attractive, which links to the wave-building recommendation.

9.2 Variation over time

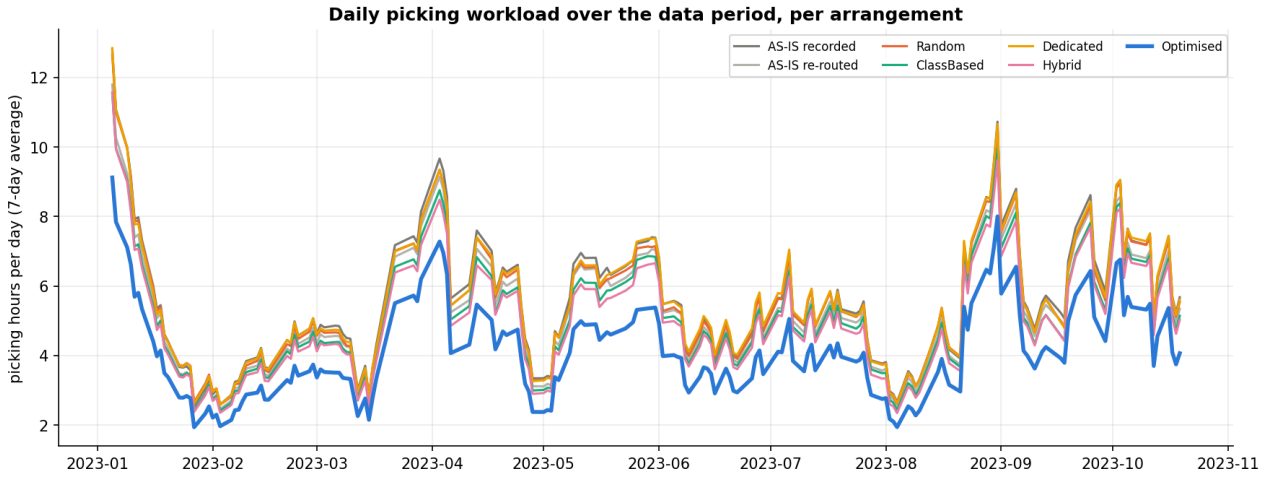


Figure 25. Daily picking workload in hours (7-day average) for every arrangement.

What it shows. Today's picking workload varies between about 2.5 and 12 hours per day, with peaks in early January, early April, late August and early October. The optimised line follows the same pattern about 25% lower.

How the code produced it. Wave times summed by release date, then smoothed with a 7-day mean (Time_Daily_Hours sheet).

What it means. The saving holds in every season, both in quiet and in busy weeks. In absolute hours the saving is largest on peak days, when it matters most.

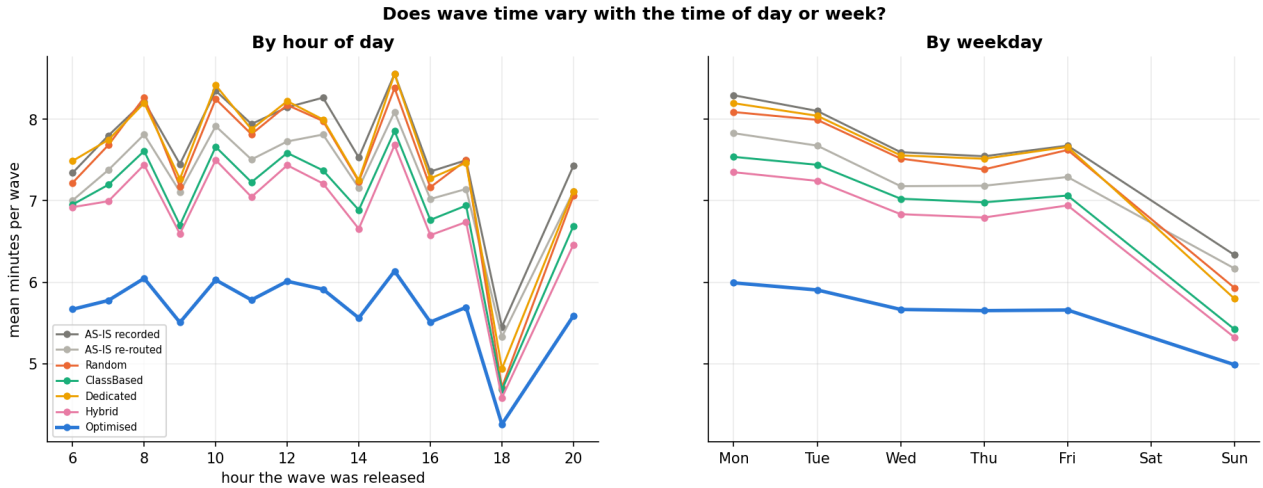


Figure 26. Mean minutes per wave by hour of release (left) and by weekday (right).

What it shows. For the current operation and case policies, mean wave time swings between about 6.6 and 8.6 min across the day (waves released at 18:00 are short, about 5 min) and falls from Monday (8.3 min) to Sunday (6.3 min). The optimised layout is flat at 5.5-6.1 min through the working day and below every other arrangement at every hour and on every day.

How the code produced it. Wave times grouped by hour and weekday of release (Time_by_Hour, Time_by_Weekday sheets).

What it means. The variation comes from what is in the waves (size, spread), not from the time of day. Because the optimised layout keeps almost every pick near the I/O, it removes most of that variation.

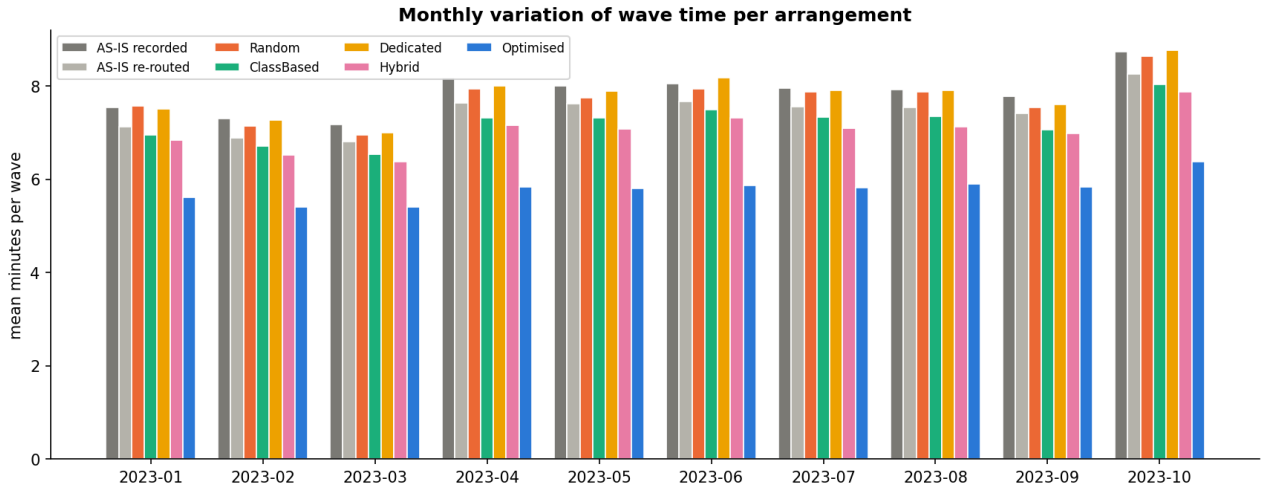


Figure 27. *Mean minutes per wave by month for every arrangement.*

What it shows. Monthly means are stable for all arrangements, with October the most demanding month (8.7 min AS-IS). The optimised layout stays between 5.4 and 6.4 min in every month.

How the code produced it. Mean wave time per release month (Time_by_Month sheet).

What it means. There is no seasonal pattern that would break the design, which is consistent with the out-of-sample result: a layout designed on January-July data keeps working in August-October.

Arrangement	Mean	Median	P10	P90	CV	Total hours	vs AS-IS
AS-IS recorded	7.85	8.06	4.42	10.95	0.32	960	-
AS-IS re-routed	7.44	7.68	4.40	10.14	0.29	910	-5.2%
Random	7.71	8.10	4.15	10.77	0.33	943	-1.8%
Class-Based	7.20	7.54	4.08	9.93	0.31	881	-8.2%
Dedicated	7.78	8.21	4.07	10.96	0.33	952	-0.8%
Hybrid	7.02	7.33	3.90	9.83	0.32	859	-10.5%
Optimised	**5.78**	**5.98**	**3.94**	**7.66**	**0.27**	**707**	**-26.3%**

Table 11. *Minutes per wave by arrangement (Time_Variation_Stats). Total hours = all comparable waves over the period.*

10. Operator performance

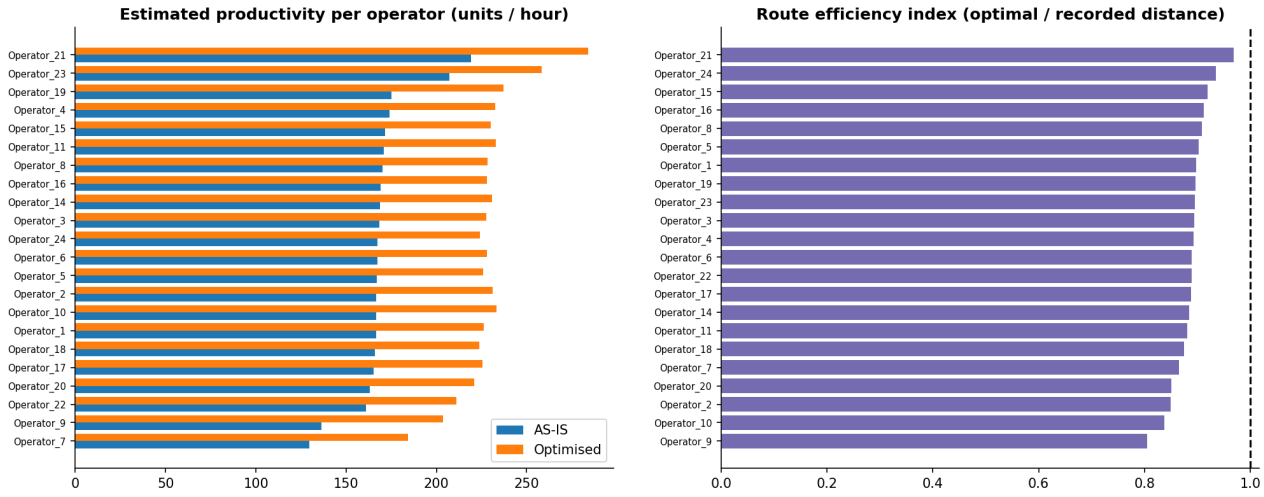


Figure 28. *Estimated productivity per operator under the current and optimised layouts (left), and route efficiency index per operator (right).*

What it shows. Left: estimated units per picking hour today range from about 130 to 220 per operator, and rise to about 185-285 with the optimised layout. Every operator gains roughly a third. **Right:** the route efficiency index divides the optimal route length by the recorded route length for each operator's waves. It ranges from 0.81 (Operator 9) to 0.97 (Operator 21); most operators sit around 0.88-0.90.

How the code produced it. Wave results grouped by the operator assigned to each wave; the index averages, per wave, AS-IS NN+2-opt distance divided by AS-IS recorded distance.

What it means. An index of 0.89 means the recorded routes are about 12% longer than necessary (1/0.89). Because the sequence comes from the WMS, differences between operators mostly reflect the waves they were given rather than individual skill. The smallest operators have very few waves, so their figures are noisy. The fix is systemic (better sequencing for everyone), not individual training.

Operator	Waves	Units	Workload share	Units/h AS-IS	Units/h optimised	Route efficiency	Hours saved
Operator_1	893	19,966	12.3%	167	226	0.90	31.7
Operator_8	809	17,866	11.0%	170	228	0.91	26.8
Operator_2	804	17,798	11.0%	167	231	0.85	29.8
Operator_17	764	16,617	10.3%	165	225	0.89	26.9
Operator_4	726	16,143	10.0%	174	233	0.89	23.5
Operator_3	653	14,591	9.0%	168	227	0.89	22.6
Operator_6	519	11,247	7.0%	167	228	0.89	17.9
Operator_14	422	9,378	5.8%	169	231	0.88	14.9
Operator_19	423	9,361	5.8%	175	237	0.90	14.0
13 other operators	1,328	28,804	17.8%	130-219	184-284	0.81-0.97	44.9

Table 12. *Operator results on comparable waves (Operator_Performance sheet).*

Workload is very unequal. Six operators handle 64% of all units, while nine operators picked fewer than 1,300 units each in the whole year (likely temporary or occasional staff). The Gini coefficient of units per operator is 0.53 (0 would mean perfectly equal). This concentration is a risk: absence of one of the six main pickers removes about a tenth of capacity. Assigning waves round-robin, and training the occasional pickers on the forward zone, would spread both workload and know-how.

11. Labour, lead time and economics

11.1 Picking labour

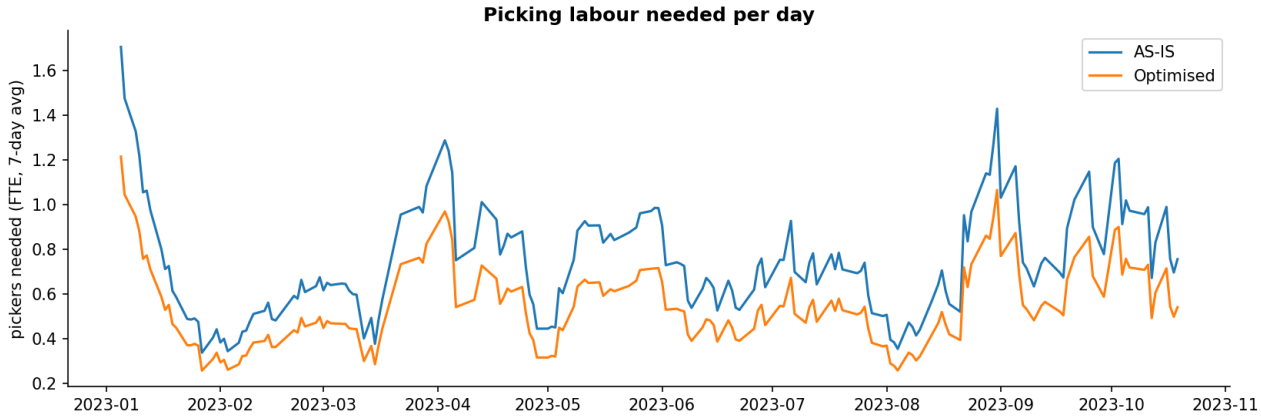


Figure 29. *Pickers needed per day (full-time equivalents of 7.5 productive picking hours, 7-day average) for the current and optimised layouts.*

What it shows. The current operation needs between about 0.35 and 1.7 FTE of pure picking time per day; the optimised layout between about 0.26 and 1.2 FTE.

How the code produced it. Daily wave hours divided by 7.5 productive hours per FTE (Staffing_Daily sheet).

What it means. The warehouse's total picking workload is small, about 5.4 picker-hours per day on average. The saving is therefore modest in head-count terms but significant in capacity: the same team can absorb peaks about a third larger before overtime or extra staff are needed.

11.2 Order lead time

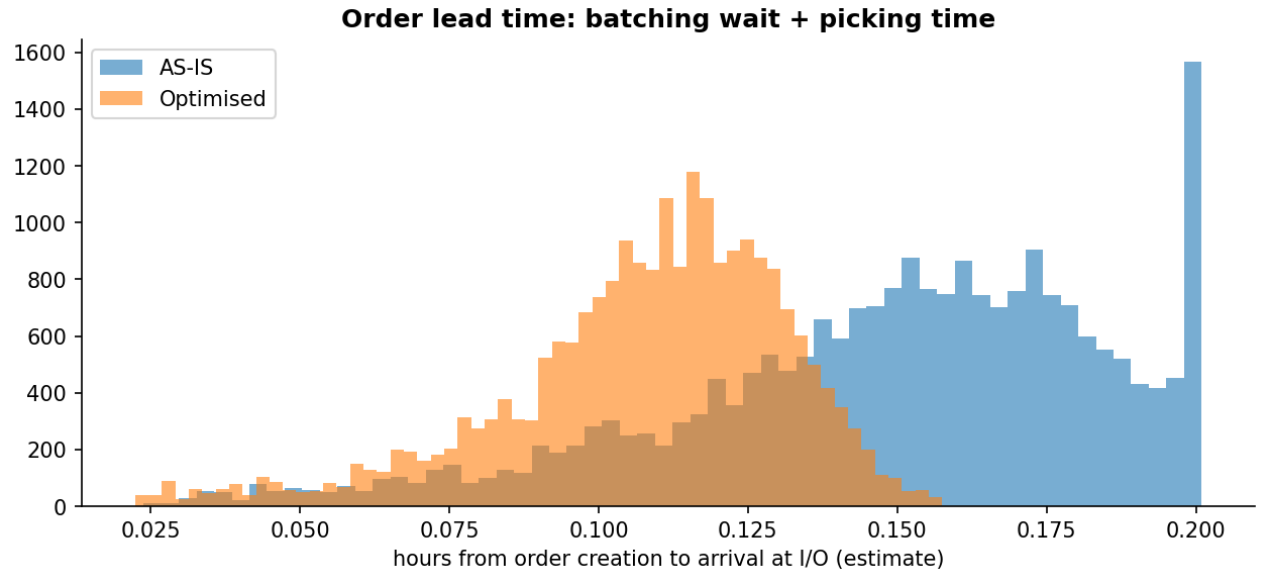


Figure 30. *Estimated time from order creation to arrival of its items at the I/O, AS-IS vs optimised (the last AS-IS bar collects all values above 0.2 h).*

What it shows. The median order reaches the I/O after 0.154 h (9.3 min) today and 0.112 h (6.7 min) with the optimised layout; the 90th percentile falls from 11.6 to 8.1 minutes.

How the code produced it. Lead time = waiting time from order creation to wave release + picking time of the wave that contains the order (Order_LeadTime_Summary sheet).

What it means. The measured waiting time is zero for every order: in this dataset the order creation timestamp coincides with the wave release. The data therefore cannot show how long orders really queue before picking, and the lead-time gain shown here comes entirely from faster picking (-28% at the median). Capturing the true order receipt time in the WMS would allow the batching opportunity seen in Figure 7 to be quantified.

11.3 Savings, sensitivity and payback

Option	Picker-hours saved per day	FTE freed	Saving per year*	Walking saved per year
Re-route only (no stock moves)	0.28 h	0.04	CAD 1,775	253 km
Re-slot + re-route (recommended)	**1.44 h**	**0.19**	**CAD 8,983**	**1,033 km**

Table 13. *Annualised savings (Savings_Annual sheet). *250 working days at CAD 25 per hour; based on comparable waves only.*

Walking speed	4 s per unit	6 s per unit (base)	10 s per unit
0.8 m/s	31.8%	29.0%	24.8%
1.0 m/s (base)	29.1%	**26.3%**	22.2%
1.2 m/s	27.0%	24.3%	20.2%

Table 14. *Sensitivity of the time saving (optimised vs AS-IS) to the two most uncertain time parameters.*

The time saving stays between 20% and 32% across the whole range of plausible parameters, so the conclusion does not depend on the exact time standards. The saving is largest when walking is slow and picking is fast, because walking then makes up a larger share of the work.

Payback. The re-slotting plan has 6,738 unit moves. At 1-2 minutes per move this is 112-225 hours of work, against a saving of 1.44 picker-hours per working day, i.e. a payback of roughly 80-155 working days. Phase 1 alone (3,818 class-A moves, 64-127 hours) captures most of the benefit and pays back faster. The monetary amounts are small because the warehouse itself is small; the main business value is faster, more predictable dispatch and spare capacity for growth, and the re-routing part is free.

12. The new layout and implementation plan

12.1 Floor plans per shelf level

The code draws each shelf level in the same style as the case layout PDFs (one labelled box per location) and colours every location by the ABC class of the demand it serves under the new layout. Vector PDFs and DXF (CAD) versions of every plan are in the outputs, so the plans can be printed or opened in AutoCAD.

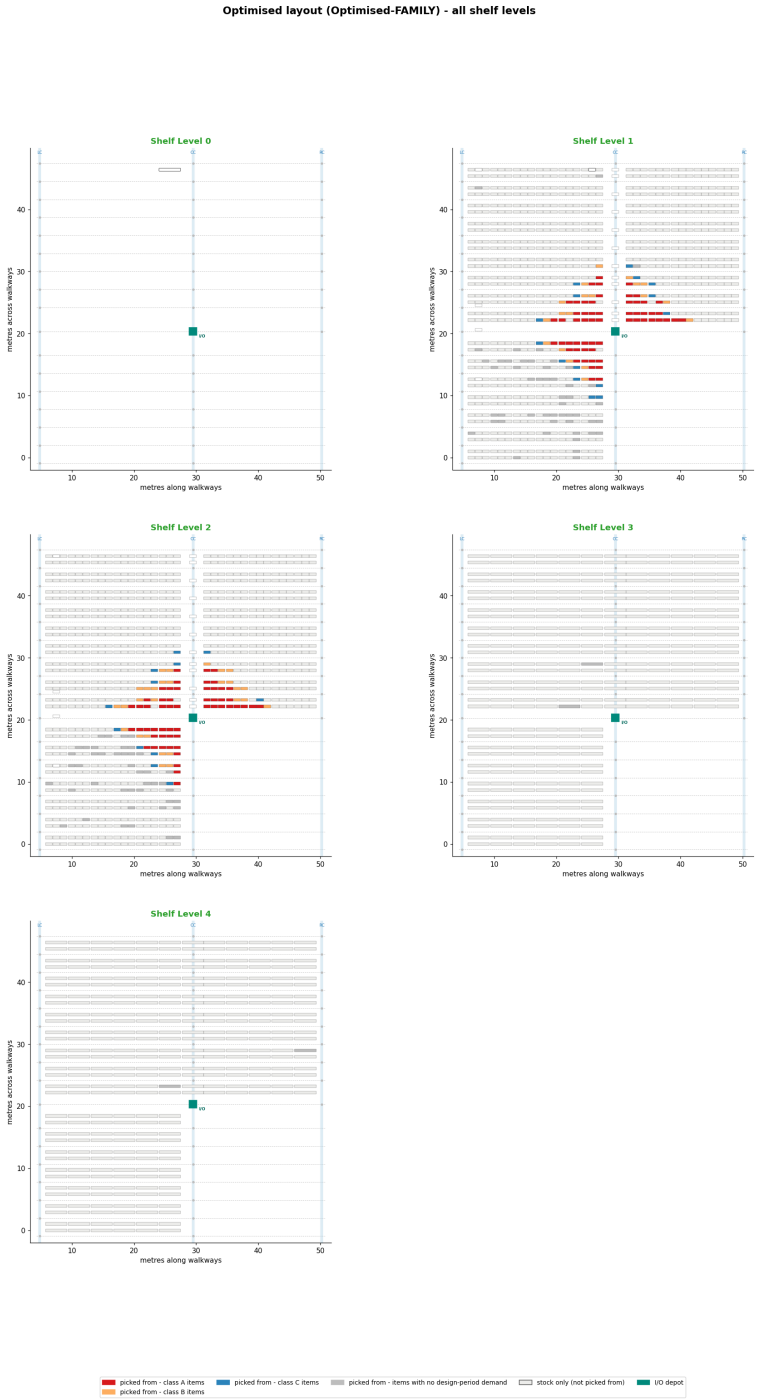


Figure 31. *Optimised-FAMILY layout, all shelf levels on one sheet (levels 0-4).*

What it shows. On levels 1 and 2 (the hand-pick levels), the class-A (red) and class-B (orange) locations form a compact block around the I/O, on walkways W08-W11 on both sides of the central corridor and on

the upper rows of the lower block next to the corridor. Levels 3 and 4 are reserve stock, so they appear as 'stock only'. 'Level 0' is not a real shelf level: it is the nine codes R-01-01 to R-01-09 whose level digit is 0, probably floor or bulk positions, drawn at one estimated position.

How the code produced it. draw_level_plans() drew one panel per level from plan_geometry() (box size from bay length and rack depth) and location_zone_classes() (class of the units picked from each location).

What it means. The forward zone occupies a small part of the rack: 186 of 2,292 locations. Everything else is a stock buffer for replenishment.



Figure 32. *Optimised layout, Shelf Level 1 (hand pick, floor height), with every location labelled.*

What it shows. Level 1 shows the forward zone in detail. The red block sits in walkways W08 (faces H and I) and W09-W10 on the right of the central corridor, and on columns 13-18 of the faces just above and below the I/O on the left. Grey locations in the lower-left block are picked occasionally by items that

had no demand during the design period.

How the code produced it. Same function, single level, with location labels from the case codes.

What it means. Each label can be matched to the New_Layout_Wide sheet, which lists the 18 boxes of each location in the same 'REF;SIZE' format as the case storage files.



Figure 33. *Optimised layout, Shelf Level 2 (hand pick, waist height).*

What it shows. Level 2 mirrors level 1, because positions on levels 1 and 2 of the same bay are the same walking distance from the I/O. The two levels together form the forward zone.

How the code produced it. Same as above.

What it means. Level 2 starts at 1.03 m, roughly waist height. A practical refinement is to put the most frequently picked and heaviest items on level 2 within each bay, which the final version of the code does automatically.



Figure 34. *Optimised layout, Shelf Level 3 (forklift replenishment).*



Figure 35. *Optimised layout, Shelf Level 4 (forklift replenishment).*

Levels 3 and 4 (Figures 34 and 35) hold one wide position per bay and serve only as replenishment stock reached by forklift. In the plan they remain unchanged; only a handful of positions are shaded, where the evaluation had to fall back on reserve stock for an item. Keeping the reserve directly above the forward zone shortens the forklift's replenishment trips.

12.2 What has to move: the re-shuffle plan

Item	Value
Baseline stock (current random storage)	41,256 boxes in 2,292 locations
Forward-zone locations re-slotted	186 (3,052 slots)
Items with a forward-zone position	1,769
Unit moves in total	6,738 (16.3% of stock) = 3,369 boxes brought in + 3,369 displaced boxes put in the vacated slots
Phase 1: class-A items and the boxes they displace	3,818 moves (1,909 swaps)
Phase 2: remaining items	2,920 moves (1,460 swaps)
Estimated effort	112-225 hours at 1-2 minutes per move

Table 15. *Re-slotting plan (02_New_Slotting_Plan.xlsx, sheets Relocation_Summary and Relocation_Moves).*

Move	Item (REF|size)	From	To	Type	Class	Phase
1	SMMRK3|11	D-14-11 slot 2	H-10-13 slot 1	bring to forward zone	A	1
2	HOUGRO|13	H-10-13 slot 1	D-14-11 slot 2	displaced to vacated slot	C	1
3	SMMRK3|11	I-23-12 slot 4	H-10-13 slot 2	bring to forward zone	A	1
4	HOUGRO|13	H-10-13 slot 2	I-23-12 slot 4	displaced to vacated slot	C	1

Table 16. *First rows of the move list: each class-A box brought forward swaps places with a slow mover.*

Item	Class	Visits (design)	Forward slots	Primary location	Distance to I/O
SMMRK3 size 10	A	458	12	H-10-13, H-10-23	2.3 m
SMMRK3 size 11	A	471	12	H-10-13	2.3 m
SMMRK3 size 10.5	A	391	10	H-10-23	2.3 m
XUZJ74 size 10	A	281	7	I-13-21	2.6 m
XUZJ74 size 9	A	222	6	I-14-21	2.6 m

Table 17. *Example rows from the Item_Plan sheet: the sizes of model SMMRK3 (recorded as class C in the master, actually the fourth most-picked reference) are grouped in one bay 2.3 m from the I/O.*

12.3 Files delivered

File	Content
01_Analysis_Tables.xlsx	32 sheets: case facts, assumptions, data quality, geometry, scenario summary, savings decomposition, statistical tests, policy profiles, operators, savings, sensitivity, staffing, lead time, all EDA tables, ABC tables, location ranking, time variation.
02_New_Slotting_Plan.xlsx	New layout in the case format (Location; ABCCOD; slots 1-18), slot-level layout with roles, item plan, relocation summary, numbered move list, design figures.
03_Reconstructed_Picking_Routes.xlsx	Per-wave summary (AS-IS vs optimised distance and time) and stop-by-stop routes for AS-IS recorded, AS-IS re-routed and the optimised layout.
figures/ and floor_plans/	All charts in this report; level-by-level floor plans as PDF and PNG.
cad/	DXF drawings of current and optimised layouts per level and all levels; open in AutoCAD and save as DWG.
tables/	CSV copies of key tables, Optimised_Storage.csv and wave-level results for all scenarios.

Table 18. *Outputs produced by one run of the code.*

13. Recommendations

#	Action	Why (evidence)	Effort
1	**Sequence pick lists with the optimised route** (nearest-neighbour + 2-opt) in the WMS.	-14.6% walking, 73% of waves improve, no stock moves (Section 8.2).	Low (IT change)
2	**Re-slot the class-A forward zone** around the I/O using the Phase 1 move list.	Most of the slotting gain with 3,818 moves (Section 12.2).	A few days
3	**Complete the forward zone** (Phase 2) and set up **daily replenishment** of it from reserve stock.	-59.5% walking, -26.3% time, +36% units/hour; robust on unseen waves (Sections 8-9).	Medium
4	**Refresh ABC classes quarterly** from WMS pick visits and re-run the slotting.	Only 56% of master classes match 2023 demand (Figure 11).	Low (run the code)
5	**Clarify RC-01** and the other non-shelf pick locations and add them to the model.	10.7% of units are picked there (Section 4).	Low
6	**Use the lower block (W01-W07) as slow-mover and overflow storage**, keeping fast movers out of it.	No picks there in 2023 (Figure 12).	Low
7	**Build waves by location proximity** for the 7 am order burst (order batching).	Orders arrive in a burst; 38% of waves serve one order; larger waves cost little extra in the new layout (Figures 7, 9, 24).	Medium (next study)
8	**Balance waves across operators** and train occasional pickers on the forward zone.	Six operators handle 64% of units; Gini 0.53 (Section 10).	Low
9	**Record scan timestamps** for each pick.	Would replace the assumed time standards with measured ones and allow the real order queue to be measured (Section 11.2).	Low

Table 19. *Recommended actions in order of implementation.*

14. Limitations and further work

- **Static evaluation.** Each pick is assigned to the nearest shelf holding the item, as if that box were always there. In reality, a shelf empties during the day and the next pick of that item must come from the next-nearest box. The forward zone in this run holds two days of *average* demand, so it would need daily replenishment and could run short on busy days. The final version of the code therefore adds a chronological stock-depletion simulation (one box per slot, overnight restocking, a forklift penalty when levels 3-4 must be used) and sizes the forward zone for a 98th-percentile day. That stricter test lowers all arrangements' results somewhat but keeps the same ranking; its numbers are not used in this report.
- **Time standards are assumed** (walking speed, time per stop, per unit and per wave) because the data has no timestamps. The sensitivity analysis shows that the conclusion holds across plausible values.
- **Product weight** is not used. The case gives three weight classes but not the weight of each product, so weight could not be modelled reliably. Shelf load limits (612 kg per level) are far above the load of 18 boot boxes and are not binding.
- **Excluded picks.** 11% of units (mainly RC-01) are not on the rack map, so 24% of waves were excluded from the comparison and the annual savings are understated by about that share.
- **Anonymised coordinates.** Distances depend on the metre calibration, which was validated against the CAD overall dimensions.

- **Size matching.** 11.3% of picks did not find their exact size in the stock file and were matched at reference level.
- **One warehouse, one year.** Results apply to this site and demand pattern; the code can be rerun on new data.
- **Further work:** order batching and wave building (Recommendation 7), replenishment scheduling for the forward zone, and correlation-based slotting (placing items that are often ordered together side by side).

15. Conclusion

The case asked how to reduce the time between receiving an order and delivering its items to the I/O point in a random-storage footwear warehouse. By rebuilding the warehouse as a walking network and replaying every 2023 picking wave under seven storage arrangements and three routing methods, this study shows that the answer lies in two complementary levers. Better routing, a pure software change, saves 14.6% of walking immediately. Demand-based slotting, which builds a compact COI forward zone around the I/O and keeps sizes of the same model together, raises the saving to 59.5% of walking and 26.3% of picking time, with 36% more units per picker-hour. It needs only 16% of the stock to move, outperforms all four storage policies supplied with the case, and works equally well on demand it was not designed on.

The analysis also turned the data into operational insights: an outdated ABC master, an unused lower rack block, a tenth of picks logged outside the rack, a 7 am order burst that invites batching, and a workload concentrated on a few operators. Because the whole pipeline is a single reproducible script, the company can rerun it every quarter on fresh WMS data to keep the layout aligned with demand.

Appendix A. Model configuration used for this report

Parameter	Value	Meaning
IO_SUPPORT_LABEL	CC-08	I/O depot at the central corridor, walkway H/I gap
AXIS_SCALE	auto	Separate metre calibration per axis
BAY_LENGTH_M / AISLE_PITCH_M	3.66 / 2.90	Calibration references from the case and CAD
PICK_LEVELS / RESERVE_LEVELS	(1, 2) / (3, 4)	Hand-pick and forklift levels
SLOTS_PER_LOCATION	18	Capacity of a location
TROLLEY_CAPACITY	27	Items per wave
WALK_SPEED_MPS	1.0	Walking speed with trolley
TIME_PER_STOP_S / TIME_PER_UNIT_S / WAVE_SETUP_S	10 / 6 / 60	Time standards
SIZE_DECIMAL_FIX	True	85 becomes 8.5
MATCH_LEVEL	auto (SKU chosen)	Exact size matched when at least 60% of picks find it
ABC_CUTS	0.80 / 0.95	Cumulative share of pick visits for A / B
SLOTTING_DEMAND	wave_visits	Items ranked by visits, not units
PICKFACE_DAYS_OF_COVER	2.0	Forward-zone stock = 2 days of average demand
TRAIN_FRACTION	0.70	Design on first 70% of waves
BASELINE_INVENTORY	Random	Stock that is re-shuffled
TWO_OPT_MAX_PASSES	10	2-opt improvement passes
LABOR_COST_PER_HOUR / WORKING_DAYS	CAD 25 / 250	Economics

Table 20. *Configuration of the run (01_Analysis_Tables.xlsx, sheet Config_Assumptions).*

Appendix B. How to reproduce the results

- Upload warehouse_optimization.py and the nine CSV files to Google Colab (folder /content/data, a Google Drive folder named warehouse_data, or the Files panel).
- Run `%run warehouse_optimization.py` in a cell. A full run takes a few minutes.
- All outputs are written to /content/outputs and downloaded as warehouse_outputs.zip.
- To reproduce exactly the static evaluation reported here with the final version of the script, set SIMULATE_DEPLETION = False, TIME_PER_KG_S = 0, LEVEL_UNIT_TIME_S = {1: 0, 2: 0} and the forward-zone sizing to 2 days of average demand.

Appendix C. Glossary

Term	Meaning
Wave	A pick list released by the WMS and collected in one trolley trip (up to 27 items).
SKU	Stock-keeping unit: a reference (boot model) in a given size.
Pick visit	One appearance of an item in a wave; the driver of walking.
ABC analysis	Classifying items by their cumulative share of demand: A = top 80%, B = next 15%, C = last 5%.
COI (Cube-per-Order Index)	Slotting rule: rank items by demand relative to the space they need and give the highest-ranked items the locations closest to the depot.
Forward zone / pick face	The hand-pick locations nearest the I/O that hold the fast movers' picking stock.
TSP	Travelling-salesman problem: the shortest tour visiting a set of points once.
2-opt	Local-search improvement for the TSP that reverses tour segments while this shortens the tour.
Serpentine / S-shape	Routing rule that traverses walkways one after another in alternating directions.
Out-of-sample test	Evaluating a design on data that was not used to build it.
Wilcoxon signed-rank test	Non-parametric paired test of whether two sets of matched values differ.
Gini coefficient	Inequality measure between 0 (all equal) and 1 (one person does everything).
FTE	Full-time equivalent; here 7.5 productive picking hours per day.

Table 21. *Glossary.*

Appendix D. References

de Assis, R.F., de Paula Ferreira, W. & Ouhimmou, M. (2025). Order picking dataset from a warehouse of a footwear manufacturing company. *Data in Brief*, 61, 111837.

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